

Pressure at Home, Presence Abroad: Development Projects and Chinese Mobility to Africa

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Abstract

Chinese-financed development projects shape more than the allocation of capital across Africa; they also serve as temporary channels for cross-border mobility. We construct mobility links between Chinese prefectures and African destinations from 76 million Weibo posts by more than 90,000 users and combine them with geocoded development projects and construction strikes in China thereby linking 295 Chinese prefectures with 1,355 African districts. When labor-market conditions in a prefecture deteriorate, Chinese mobility in Africa shifts toward locations with ongoing Chinese-financed projects, while total mobility from affected prefectures to Africa remains unchanged. The shift begins one to three months before a construction strike is recorded. At World Bank-financed project sites, mobility shifts only after the strike. These patterns suggest that ongoing development projects temporarily connect particular African locations to changing economic conditions in China by providing channels through which cross-border mobility can adjust.

Keywords: Africa, China, development projects, international mobility, shift-share designs, social media

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1 Introduction

Chinese-financed development projects have become an important part of the economic landscape of many African countries. Between 2000 and 2023, Chinese lenders committed 1,306 loans worth \$182 billion to 49 African governments, financing infrastructure and other projects across the continent (Boston University Global Development Policy Center, 2024; Gelpert et al., 2023). Much of this financing is tied to Chinese contractors. As a result, a project does not only bring capital to an African location; while it is under implementation, it also brings a foreign firm, organizational infrastructure, and a recurring movement of people between China and the African project site (Brautigam, 2009; Oya and Schaefer, 2019). Yet little is known about how these project-based connections shape the geography of international mobility into Africa or how responsive they are to economic conditions in the country supplying the firms and personnel.

This paper studies whether ongoing development projects connect particular African locations to labor-market conditions in specific parts of China. We ask whether deteriorating economic conditions in a Chinese prefecture shift Chinese mobility within Africa toward locations where Chinese-financed projects are already under implementation. The question matters for understanding the broader economic footprint of development finance in Africa. Foreign-financed projects are usually studied through their effects on capital allocation, infrastructure, local economic activity, or employment. But implementation also establishes organizational links between recipient locations and firms, workers, subcontractors, and suppliers abroad. These links may temporarily integrate otherwise weakly connected African locations into international production and mobility networks.

To explain these connections, we build on migration models in which established networks lower the fixed costs of reaching particular destinations. Conventional migrant networks develop gradually as earlier migrants provide information, referrals, and housing (Munshi, 2003; Beaman, 2012; Bryan, Chowdhury and Mobarak, 2014). An ongoing development project can create a related form of capacity much more quickly. A firm already operating at an African project site can provide employment opportunities, accommodation, transport, administrative support, and an established route to a particular location. We call this *placement capacity*. When labor-market conditions deteriorate in China, this capacity can make some African destinations easier to reach than others. The model therefore predicts not necessarily an increase in total mobility to Africa, but a reallocation of Africa-bound mobility toward locations where such capacity already exists. Unlike conventional migrant networks, project-based placement capacity arises with implementation and weakens when the project ends.

We test this prediction using a new micro-level dataset that links Chinese origins to destinations across Africa. We combine 76.3 million Weibo posts from 93,971 users with geocoded Chinese- and World Bank-financed development projects and construction strikes in China. From the posting histories we construct observed stays in African countries and assign users to Chinese prefectures using their profile registration locations. Our main analysis covers 17,692 mobility

corridors connecting 295 Chinese prefectures to 1,355 African destinations between 2014 and 2019. This allows us to observe connections between specific Chinese labor markets and individual African locations at a level of geographic detail that is unavailable in conventional migration statistics.

Our main empirical design compares, within the same Chinese prefecture and year, mobility toward African destinations where a Chinese-financed project is already underway with mobility toward that prefecture’s other African destinations. Pair fixed effects absorb persistent bilateral connections, origin-by-year fixed effects absorb total observed Africa-bound mobility from each Chinese prefecture, and destination-by-year fixed effects absorb conditions common to each African location. Identification therefore comes from changes in the distribution of a prefecture’s mobility across African destinations rather than from differences in which African locations receive projects.

We find that worsening labor-market conditions in China change where Chinese mobility appears within Africa. In the year before a construction strike is recorded, one additional future strike is associated with 0.52 percent more users and 0.74 percent more new mobility spells at African destinations with an ongoing Chinese-financed project, relative to other African destinations reached from the same prefecture. At monthly frequency, the timing is sharper: one to three months before a strike is recorded, ongoing-project destinations receive 0.85 percent more users and 0.82 percent more new spell starts. We find no comparable increase in the strike month or during the following twelve months. Total observed mobility from the affected prefecture to Africa does not measurably increase. The response is therefore a redistribution of mobility across African locations rather than a rise in overall Africa-bound mobility.

The timing is consistent with construction strikes marking the visible culmination of labor-market deterioration rather than its onset (Lee, 2007; Elfstrom, 2021). Three additional results connect the reallocation to project implementation. First, the pre-strike response is larger where the implementing firm is based in the affected Chinese prefecture: the increase in new mobility spells is 1.44 percent at such project locations, compared with 0.62 percent elsewhere. Second, the timing differs at World Bank-financed project locations. Chinese-financed sites show a pre-strike increase, whereas World Bank locations show increases only after the strike. This is consistent with Chinese-financed projects being more likely to involve firms with regionally concentrated recruitment and subcontracting networks, while competitively procured projects are more often implemented by firms recruiting nationally. Third, the pre-strike differential disappears once the Chinese-financed project has been completed. Together, these patterns suggest that implementation itself creates temporary connections between particular African locations and particular Chinese labor markets.

The paper contributes to research on development finance in Africa by shifting attention from where projects are allocated to the connections that exist while they are being implemented. Existing work has studied the geography, terms, and economic consequences of Chinese development finance (Dreher et al., 2019, 2022; Bluhm et al., 2025; Horn, Reinhart and Trebesch, 2021; Gelpert et al., 2023), as well as employment and labor relations around Chinese firms and projects in Africa (Cervellati et al., 2022; An, Guo and Jiang, 2025; Lee, 2017; Fei, 2020; Oya and Schaefer, 2023;

Yang, 2022). We show that project implementation also creates temporary origin–destination links through which economic conditions outside Africa can affect where foreign mobility appears within the continent. This adds an origin-side dimension to the study of foreign economic activity in Africa: African project locations are not only recipients of capital and firms, but nodes in cross-border networks whose use responds to conditions in the economies to which those firms are connected.

The paper also contributes to research on migration networks and international mobility. Migrant networks reduce the costs of reaching particular destinations by providing information, referrals, and housing, but they accumulate gradually (Munshi, 2003; McKenzie and Rapoport, 2010; Beaman, 2012). Formal placement channels can overcome similar mobility frictions without requiring a pre-existing migrant network (Mobarak, Sharif and Shrestha, 2023; Gibson and McKenzie, 2014). We identify development projects as another such channel. Their distinctive feature is geographic and temporal specificity: a project creates organizational capacity at a particular African location while implementation is underway. This helps explain why mobility can respond to changing conditions in China without requiring either a new project award or a persistent migration network.

Finally, the paper contributes new evidence on Chinese mobility within Africa. Existing statistics are highly aggregated, and even the official count of licensed contract workers does not capture the broader population of Chinese nationals moving across the continent. Digital traces allow us to recover origin–destination links at much finer geographic and temporal resolution. Rather than treating Africa as a single destination, we observe how mobility is distributed across more than one thousand African locations and how that distribution changes with project activity. This makes it possible to study development projects as temporary links in the economic geography connecting African locations to the rest of the world.

The rest of the paper is organized as follows. Section 2 describes the institutional setting, the structure of Chinese development finance in Africa, and project staffing. Section 3 develops the conceptual framework. Section 4 describes the data and measurement. Section 5 presents the empirical strategy. Section 6 reports the results, Section 7 discusses interpretation and limitations, and Section 8 concludes by summarizing the main findings and their broader implications.

2 Context

2.1 Chinese Projects and Origin–Destination Links

Chinese development finance typically reaches Africa as tied lending, bringing Chinese implementing firms with it. The financing is mainly in the form of loans rather than grant aid, extended by state institutions on commercial or near-commercial terms outside the OECD-DAC aid category (Brautigam, 2009; Dreher et al., 2022; Horn, Reinhart and Trebesch, 2021; Gelpert et al., 2023). Between 2000 and 2023, Chinese lenders committed an estimated 1,306 loans worth \$182 billion to 49 African governments, with China Eximbank and China Development Bank accounting for

roughly 83 percent of the volume (Boston University Global Development Policy Center, 2024). Commitments accelerated in the early Belt and Road years and peaked between 2013 and 2018, so our estimation window covers the years in which the largest number of projects were under implementation (Chen, Fornino and Rawlings, 2024). Concessional and preferential credits are typically conditioned on the use of Chinese contractors, in most cases large state-owned construction groups (Brautigam, 2009; Gelpert et al., 2023). A project under implementation therefore means a Chinese contractor with an established presence at the site and a standing reason to move personnel there.

Contrary to long-standing Western narratives suggesting that Chinese firms import all the staff they need, implementing firms typically run their projects with a small dispatched Chinese cadre alongside a much larger local workforce. Around 85 percent of project employees are African, and a survey of more than 1,000 Chinese firms in eight countries finds 89 percent African employees but only 44 percent African managers (Sautman and Yan, 2015; Sun, Jayaram and Kassiri, 2017; Oya and Schaefer, 2021). The employment of local employees in Chinese firms also varies across sectors, where projects with advanced technology and management requirements such as hydropower and telecommunication normally hire more Chinese compared with projects or firms that are more labor-intensive (Xiaoyang, 2016). The Chinese presence per project is therefore modest in headcount, concentrated in management, technical, and skilled positions, and front-loaded in the project cycle (Cooke, Wang and Wang, 2018; Lee, 2017; Yang, 2021). Chinese statistics distinguish between contracted project workers and labor-service cooperation. The official worker stock in Africa peaked at roughly 264,000 in 2015 and excludes informal migrants (China Africa Research Initiative, 2025). In addition to starting new projects, firms can adjust the number of workers they employ on existing projects. Rotations can be lengthened or accelerated, or specialists can be added without a new contract award, thereby increasing the number of people observed at an already active site rather than opening a new one.

Dispatched personnel are not confined to construction projects. Chinese medical teams have rotated through Africa since 1963, sending more than 20,000 professionals to 51 countries in units of typically 20 to 30 members, and each Chinese province is twinned with specific African countries. Hubei province has staffed teams in Algeria since the first mission, Hebei province has staffed teams in the DRC since 1973, and Hunan has staffed teams in Sierra Leone and Zimbabwe (Chen et al., 2019; Shi, 2016). The 23 Agricultural Technology Demonstration Centers built since 2006 are likewise staffed by Chinese agronomists on assignment (Brautigam and Tang, 2009; Xu et al., 2016). Such twinning arrangements illustrate one possible source of persistent origin–destination links. Time-invariant links of this kind are absorbed by our origin–destination-pair fixed effects, but we do not observe a comprehensive set of province–country twinning arrangements.

While many policy discussions focus on dispatched workers, they constitute only a minority of the Chinese nationals present in Africa. A larger population of entrepreneurs, traders, and shopkeepers is documented qualitatively across the continent, while a smaller flow of tourists is concentrated

in a handful of destinations such as Egypt, Morocco, and Mauritius (Mohan and Lampert, 2013; Lampert and Mohan, 2014). Aggregate stock estimates in circulation, including the frequently cited one million, are journalistic estimates without an official basis (French, 2014). Chinese nationals in Africa include not only dispatched project personnel but also entrepreneurs, traders, shopkeepers, and tourists. Our Weibo data capture these populations without allowing us to identify occupation or reason for travel from geolocated posts. The estimates should therefore be interpreted as changes in observed Chinese mobility and presence rather than as changes in project-worker headcounts.¹

An ongoing project nevertheless creates a distinctive link to a particular African location. The implementing firm is already present, personnel and subcontracting arrangements are in place, and recruitment can draw on an operation that already exists abroad. From the perspective of the African destination, project implementation therefore temporarily embeds the location in the firm’s wider organizational and labor network. This makes adjustment through an active project fundamentally different from adjustment that first requires a new project to be awarded.

2.2 Adjustment along Existing Project Links

Firms can respond to labor-market distress at home by staffing projects they are already running, without waiting for a new contract. Distress may also change which prefectures win new contracts. Mueller (2025), for example, shows that unrest in a Chinese prefecture predicts subsequent foreign-aid contracts for firms from that prefecture. But firms can also adjust through projects that are already under implementation, without waiting for a new contract to be awarded.

Firms and local governments can also work through projects that are already running. Both responses can occur simultaneously in the same prefecture in the same year, but they differ in actor and the type of information acted upon. Separating the second requires excluding projects that begin in the year of the shock, since a differential there would be consistent with an award rather than an existing capacity.

The second distinction concerns how a change in mobility becomes visible. A destination can record more people than before, or it can record people for the first time. An adjustment that works through existing capacity should appear mainly in the first form, since the link between a prefecture and a project location already exists and the mechanism adds to its use. The second form arises when an expanded operation reaches a location to which a prefecture had not previously sent anyone. Since both may occur, our design must be able to register either one.

The third distinction is timing. Recorded labor unrest is typically the visible culmination of a deterioration that has developed over time. In Chinese construction, wage arrears and related disputes can accumulate for months before workers engage in collective action (Lee, 2007; Elfstrom, 2021). A recorded strike is therefore better interpreted as the point at which underlying labor-market

¹The content of posts, broader movement patterns, and specific destinations linked to tourism are margins for identification that we are in the process of testing.

distress becomes publicly observable than as its onset.

Firms with recruitment ties to a prefecture need not wait for that public signal. Through their relationships with workers, subcontractors, and local employers, they may observe worsening conditions and growing labor availability before a strike occurs and adjust recruitment or staffing at an existing African project site. Where such recruitment links are weak or absent, the strike itself can reveal a newly available pool of workers. This distinction implies that mobility may precede the recorded strike where firms are already connected to the affected prefecture, but follow it where those connections are weaker.

Such an anticipatory response need not prevent the strike itself. Overseas opportunities are small relative to local labor markets and may reach only particular occupations or workers. The same underlying deterioration can therefore generate both some outward mobility and, subsequently, a recorded strike.

3 Conceptual Framework

We use a random-utility model of destination choice to underpin our main empirical specifications and to derive the timing of the response from an explicit assumption about who observes labor-market conditions at the origin. The model builds on standard migration economics in two respects. Bilateral migration costs include a sunk pair-specific component and a component that migrant networks reduce. A worker chooses between moving to an African destination and a composite outside option that includes staying in China, moving elsewhere within China, or going to another part of the world. Among African destinations, moving is less costly where a migrant network or an active project already provides connections, information, and other forms of support. We therefore allow active project sites to reduce bilateral mobility costs in a way similar to migrant networks, but without requiring those networks to accumulate over time. Labor-market distress at the origin also makes workers more available for recruitment, giving firms an incentive to draw more heavily on origins where conditions have deteriorated. The model shows how these mechanisms affect where workers go and yields the predictions we test in the data.

3.1 Destination choice and placement capacity

A worker i in Chinese prefecture o chooses in period t between an African destination $d \in D$ and a composite outside option that covers staying in o , moving elsewhere within China, and moving to a region outside Africa. The choice depends on four objects. The return w_{dt} available at destination d varies across destinations and over time but is common to workers from different Chinese origins. The bilateral cost c_{odt} is what a worker from o pays to reach d and take up an opportunity there. The value u_{ot} is what the outside option is worth to workers in o at time t . The terms ε_{iodt} and ε_{iot}^0 capture worker-level differences we do not observe, such as individual preferences, personal contacts,

or occupation-specific opportunities. Payoffs are

$$V_{iodt} = w_{dt} - c_{odt} + \varepsilon_{iodt}, \quad V_{iot}^0 = u_{ot} + \varepsilon_{iot}^0, \quad (1)$$

where the unobserved components follow a generalized extreme value distribution with two nests, the set of African destinations D and the outside option, and dissimilarity parameter λ .²

Labor-market distress in the origin, denoted ξ_{ot} , lowers u_{ot} through the expected wage at home and lowers the wage at which firms can recruit workers from o . Because u_{ot} is a composite, it falls by less than the value of staying does. Internal migration is the cheapest adjustment available to a rural migrant working in domestic construction, and it sits inside the outside option rather than in D .

The bilateral cost has three parts,

$$c_{odt} = \kappa_{dt} + F_{od} - \phi \ln A_{odt}, \quad \phi > 0. \quad (2)$$

The term κ_{dt} collects destination conditions that affect workers from all Chinese origins similarly, such as transport access, local prices, regulation, and political conditions. The term F_{od} is the sunk cost of establishing a corridor between o and d at all. It does not vary with short-run labor-market conditions once the corridor exists. Fixed costs of this kind bind in models and evidence on migration (Bryan, Chowdhury and Mobarak, 2014; Lagakos, Mobarak and Waugh, 2023; Clemens, Montenegro and Pritchett, 2019). The term A_{odt} is placement capacity. It is the stock of arrangements at d that lowers the cost to a worker from o of using a corridor whose fixed cost has already been paid, and ϕ determines how strongly it does so. These arrangements can include a job offered without a search, accommodation on arrival, and an entry permit filed by someone else. A firm running a project at d can supply all three, so we let placement capacity have two components,

$$A_{odt} = N_{odt} + \omega_{od} K_{dt} g(I_{odt}). \quad (3)$$

The first component, N_{odt} , is the migrant network from o at d . It is a slow-moving stock accumulated by past flows, and it lowers costs by supplying job information, referrals, and housing on arrival (Munshi, 2003; McKenzie and Rapoport, 2010). Because such networks are built through previous migration, N_{odt} responds to short-run conditions at the origin only gradually.

The second component is project-based. The term K_{dt} is the capacity of active project sites at d . It is positive while a project is under implementation, when an implementing firm, a worksite,

²The dissimilarity parameter governs how substitutable African destinations are for one another relative to the outside option. When one destination becomes less attractive, a low λ sends most of the displaced mobility to other African destinations rather than to China or the rest of the world, because the unobserved reasons for choosing any African destination largely overlap. At $\lambda = 1$ the model collapses to conditional logit, and any change within Africa spills onto the leave-or-stay margin in proportion to existing shares. We do not estimate λ . It enters θ in equation (6) as a scale factor and does not affect its sign.

and staffing arrangements are already in place, and it falls when the project is completed. Project capacity at d is not equally useful to workers from every Chinese origin. The parameter ω_{od} scales how useful such a site is to workers from o . It is larger when a contractor at d recruits from o , has subcontractors or suppliers there, or maintains other origin-specific links, and we treat it as time-invariant over the estimation period. The function $g(\cdot)$ instead varies with conditions at the origin. It gives the intensity with which site operators recruit from o and increases in the information I_{odt} they hold about labor availability there. Operators connected to o observe deteriorating conditions as they develop, so we let I_{odt} track ξ_{ot} one for one and write g' for the derivative of recruitment intensity with respect to ξ_{ot} .

An active site supplies each of these arrangements. It offers a hiring channel through the implementing firm, accommodation and transport, and legal entry through a licensed labor-cooperation firm (State Council of the People's Republic of China, 2012). It supplies them as soon as implementation begins. A migrant network takes longer to reach the same point, because only settled earlier migrants can pass on job information and referrals (Beaman, 2012). Formal placement channels are known to reach workers who lack networks abroad (Mobarak, Sharif and Shrestha, 2023; Gibson and McKenzie, 2014). Treating a discrete, time-limited project as one such channel is, to our knowledge, new.

Equation (3) makes N_{odt} and K_{dt} substitutes, which we assume rather than estimate. A worker who can already learn about a job through relatives or friends at the destination gains little from a site that offers the same thing, while a worker with no such contacts gains a great deal. The marginal value of a project is therefore lower where the migrant network is thick. The two components also differ in what they leave behind. Project capacity ends with the project, whereas the connections created by earlier movers can persist after the site closes.

3.2 Site operators

Project capacity at destination d lowers mobility costs for workers from origin o only if firms operating at the site recruit from that origin. We therefore distinguish between the capacity available at a project site and the origin-specific recruitment links through which firms use it. These links determine both which Chinese prefectures are connected to a given African project site and when firms respond to changing labor-market conditions in those prefectures. It also underlies the donor comparison in Section 5.3.

Firms are more likely to recruit from origins where they already have established relationships with workers, subcontractors, or local partners. Where these arrangements connect the operator of a project at destination d to prefecture o , so that ω_{od} is large, the firm can observe deteriorating labor-market conditions there as they develop and adjust recruitment before those conditions culminate in a recorded strike. Where such links are weak or absent, workers from o are more likely to reach the site only after the strike makes the deterioration and the resulting pool of available

workers publicly visible. These origin-specific recruitment links are likely to differ across financing arrangements. Chinese-financed contracts are allocated through MOFCOM to firms anchored in particular provinces, an allocation channel that Mueller (2025) documents at the prefecture level. Competitively procured civil works, including those financed by the World Bank, are more often implemented by central state-owned enterprises with national recruitment networks (Foster et al., 2009). The first type of recruitment link should therefore be more common at Chinese-financed sites, while the second should be more common at World Bank-financed sites. This is a probabilistic distinction rather than a strict partition of contractors by financing source, and we examine it directly using contract-winner data.

That firms recruit disproportionately from their own province or prefecture is an institutional hypothesis. Chinese construction firms internationalize as networks of provincial origin, with subcontractors and suppliers of shared provenance following the lead contractor abroad (Fei, 2021). The hometown ties through which domestic construction labor is recruited are reproduced on overseas sites (Fei, 2020). No source we are aware of documents recruitment for a specific overseas project by prefecture. We treat this as the weakest step in the argument and test it directly in Section 5.3.

New contract awards constitute a third margin. They are decided centrally. They create capacity that does not yet exist and cannot produce a response at sites already running. Our baseline project definition excludes them.

3.3 Distress, thresholds, and the recorded strike

Labor-market distress need not begin when a strike is recorded. In Chinese construction, wage arrears and related grievances often build over several months before workers engage in collective action (Lee, 2007; Elfstrom, 2021). We capture this by letting ξ_{om} denote the additional distress arising in prefecture o in month m , and by defining accumulated distress as

$$\Theta_{om} = \sum_{k \leq m} \xi_{ok}.$$

The variable Θ_{om} therefore summarizes how far deteriorating conditions have progressed by month m .

We assume that firms and workers respond to this accumulated distress at different thresholds. For a firm already recruiting in prefecture o , adjusting staffing at an African project site involves fixed costs. Organizing travel, permits, and worker handovers is costly even for a small adjustment, so firms have little reason to change staffing continuously in response to small changes in local conditions. Instead, the operator waits until worsening labor-market conditions make recruitment sufficiently attractive to justify these adjustment costs. We denote this firm threshold by τ_F . Once

$$\Theta_{om} \geq \tau_F,$$

the operator adjusts recruitment or staffing at the African site.

Workers face a different threshold. Collective action is itself costly and requires sufficiently severe and widespread grievances before workers are willing to participate. We denote this strike threshold by τ_S , and assume that

$$\tau_F < \tau_S.$$

Accumulated distress can therefore become large enough for a locally connected firm to adjust recruitment before it becomes large enough to generate a recorded strike. When Θ_{om} eventually crosses τ_S , collective action occurs and the deterioration becomes publicly visible.

This ordering generates the timing difference that is central to the empirical analysis. A site operator with recruitment links to prefecture o can respond when Θ_{om} crosses τ_F , so mobility toward that site can increase before a strike is recorded. An operator without such links does not observe the deterioration as directly. For these firms, the recorded strike provides a public signal that workers in the prefecture have become more available, so the mobility response can occur only after the strike.

We denote by L_{om} the number of strikes recorded in prefecture o in month m . Recorded strikes are compiled from worker and media reports and therefore capture only a subset of actual labor disputes. We consequently use L_{om} as an observable indicator of underlying distress rather than as a measure of its absolute level. Variation in recorded strikes identifies differences in the incidence and timing of labor-market deterioration across prefectures and periods, while the incomplete reporting of strikes attenuates the link between observed L_{om} and the underlying process.

3.4 From the model to the estimating equation

Let M_{ot} denote the number of potential migrants from o in period t , scaled by the probability that a mover from o is observed on the platform. The expected number of users we observe from o at d is that number times the share choosing the African nest times the share choosing d within it,

$$E[Y_{odt}] = M_{ot} P(D | o, t) P(d | D, o, t), \quad P(d | D, o, t) = \frac{\exp\{(w_{dt} - c_{odt})/\lambda\}}{\sum_{d' \in D} \exp\{(w_{d't} - c_{od't})/\lambda\}}. \quad (4)$$

Take logs and substitute (2). Everything that varies only by origin and period is absorbed by the origin-by-year effect η_{ot} of equation (8), which covers M_{ot} , the nest share, and the log-sum. The destination-by-year effect μ_{dt} absorbs $(w_{dt} - \kappa_{dt})/\lambda$, and the pair fixed effect α_{od} absorbs $-F_{od}/\lambda$. What remains is $(\phi/\lambda) \ln A_{odt}$. Expanding (3) around the pair mean $\bar{A}_{od}$, with $\bar{\xi}_o$ the average level

of distress in o , gives

$$\frac{\phi}{\lambda} \ln A_{odt} \approx \frac{\phi}{\lambda} \ln \bar{A}_{od} + \frac{\phi}{\lambda} \frac{\omega_{od} g'}{\bar{A}_{od}} K_{dt} (\xi_{ot} - \bar{\xi}_o). \quad (5)$$

The first term is a pair fixed effect. In the second, K_{dt} on its own is absorbed by the destination-period fixed effect, and ξ_{ot} on its own by the origin-period fixed effect. Their interaction survives all three sets of fixed effects. Writing $\rho = \partial \xi / \partial L$ for the mapping from the recorded strike count to distress at the time mobility is measured, the surviving object is

$$\theta = \frac{\phi}{\lambda} \frac{\omega_{od} g'}{\bar{A}_{od}} \rho. \quad (6)$$

Equation (6) shows four things. First, θ captures where migrants go rather than how many leave. Total mobility from o depends on M_{ot} and the share choosing Africa, and the origin-period fixed effect absorbs both. This is the reallocation object discussed by Bertoli and Fernández-Huertas Moraga (2013). Second, θ rises with ω_{od} . Thus, the estimate is larger for pairs in which the firms operating at d recruit in o . Third, θ falls with $\bar{A}_{od}$, so project capacity matters less where the migrant network is already thick. Fourth, the timing of the response is governed by ρ . Where operators recruit in o , distress is elevated before the strike, and ρ is positive in the pre-strike window. Where they do not, workers arrive only afterward, and ρ is positive in the post-strike window.

Tourists and traders are in M_{ot} and do not respond to project capacity, so $g' = 0$ for them. Poisson pseudo-maximum likelihood identifies a proportional effect, so their presence in the same pairs scales θ toward zero by their share. Estimates for spells confined to a single administrative unit, a behavioral proxy for project-based presence, therefore bound the effect among responsive users rather than only contrasting travel types.

3.5 Predictions

Composition responds. The model predicts a positive θ , the shift of an origin's mobility toward destinations with a project already running. It imposes no restriction on the size of the response of total mobility from o . Three features of the setting suggest that the volume response is small. Headcount at running sites is determined by the project pipeline, so labor market distress at origin affects who fills the available positions and how quickly, not how many positions there are. For a construction worker facing arrears at home, the cheapest margin of adjustment is internal migration, which we place in the outside option. Furthermore, the requirements of going abroad at all, from licensed dispatch to passports and visas, impose a cost that is likely to matter for the marginally exposed worker. Consistent with this, the origin-level estimates in Section 6 take the same sign as the bilateral ones but are an order of magnitude smaller and indistinguishable from zero.

The response occurs at sites already running. K_{dt} counts the placement a site can supply, so it is positive only where a site exists. This separates the margin we study from the allocation of new

projects, as documented by Mueller (2025).

Timing differs by operator. Where operators recruit in o , they adjust once Θ_{om} crosses τ_F , so mobility leads the recorded strike. Where they do not, workers from o arrive only after τ_S is crossed, so mobility lags it. The distinction is not necessarily between Chinese- and non-Chinese-financed sites. Chinese firms build a substantial share of multilateral civil works in Africa (Foster et al., 2009), so a World Bank site may be staffed by Chinese personnel and still have no recruitment link to the prefecture under distress. What varies with the financing source is the probability of that link. Chinese-financed contracts are allocated through restricted or open tendering organized by MOFCOM to firms anchored in particular provinces,³ while competitively procured civil works, including those financed by the World Bank, are more often implemented by central state-owned enterprises with national recruitment networks (Foster et al., 2009).⁴ A recruitment link to the prefecture under distress should therefore be more likely at Chinese-financed sites. Chinese-financed sites should consequently display the lead, World Bank sites the lag.

The response requires an active site. Placement capacity K_{dt} , the placement that active project sites at d can supply, falls to zero at project completion. The camp closes, the payroll ends, and the firm that recruited into it moves on. Destination conditions κ_{dt} do not change, since the port, mine, or city that makes a location attractive remains in place. A location that draws mobility because a project runs there should therefore stop showing a strike differential once the project ends, while a location that draws mobility for its own reasons should be unaffected by completion. The prediction applies to both timing patterns. Where operators recruit in the prefecture under distress, the differential that disappears is the one in the months before the strike. Where they do not, it is the difference in the months that follow. Whether a completed project leaves a network behind is a distinct question, about the level of presence at the location rather than about its response to distress at a particular origin, and it calls for a design around completion dates rather than a strike differential.

Project capacity matters more where the network is thin. Average placement capacity $\bar{A}_{od}$ combines the migrant network N_{odt} with what project sites supply. If the two are substitutes, a worker who could already learn about a job through relatives or friends at the destination gains little from a project site that offers the same thing, whereas a worker with no such contacts gains a great deal. The marginal value of a project is then lower where the network is thick, which is what $\bar{A}_{od}$ in the denominator of (6) delivers. A direct test of this prediction would require an independently measured origin–destination network and an interaction between network strength, labor-market distress, and active-project status. We do not observe a sufficiently comprehensive external measure of such networks. We nevertheless examine a related implication using origin-specific destination shares measured before the main estimation period. This exercise provides suggestive evidence on

³See https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzbmzg/201203/t20120328_145137.html.

⁴Private firms, which generally have weaker political backing, financial capacity, and risk-bearing capabilities, rarely secure large-scale overseas aid-funded infrastructure contracts directly from the Chinese government. Instead, they are more likely to participate as subcontractors to SOEs.

whether mobility responses are stronger along historically thin corridors, but it does not isolate whether the additional effect of an active project is itself larger in those corridors.

Three findings would be inconsistent with the mechanism we laid out: A lead at sites whose operators do not recruit in the shocked prefecture, World Bank sites included, would contradict the timing structure. A differential concentrated in pairs lacking such a recruitment link would contradict the placement channel. And finally, a differential that survives project completion would point to location characteristics rather than capacity.

4 Data and Measurement

4.1 Weibo posts and African destinations

Weibo is a Chinese microblogging platform operated by Sina Corporation and one of the largest social networks in China, with more than 510 million monthly active users by September 2020 (Weibo, 2021). Each post record contains the user identifier, the post date, coordinates (where present), the post text, and counts of replies, comments, and likes. Some posts also contain a location, either as coordinates or as a named point of interest such as a sight, a hotel, a shop, or an administrative area. Such tags allow geo-locating the posts.

In addition to the post’s data, there is information from user profiles. Profile records hold a user identifier, nickname, gender, user level, the registration location, education and work history, and counts of followers and friends. The registration location can be used to identify the origin province of users.⁵ We work only with pseudonymized data.

Sampling proceeded in two stages. In 2020, we randomly sampled 10 million users from the platform’s user base and scraped their complete posting histories back to account registration; 90,169 of them had posted at least once from a location in Africa. In 2022, we added 3,089 users found by searching for African points of interest in the Weibo application from 2018 to 2022. For every user in the combined set we then scraped all historical posts, geotagged or not, through the end of 2022, along with profile characteristics. Of these users, 3,333 have contributed no posts, either because their account access is restricted or because they deleted their posting history.

The two draw dates shape which accounts we observe. The first sample consists of accounts that were alive in 2020, so the further back the panel runs, the more it is built from long-lived users. The second draw reaches accounts the first one missed, and because we scraped the full history of every user in that set, it also contributes posts from before its search window. The composition of observed accounts therefore changes over the sample period. The origin-by-year fixed effects hold that composition fixed within each prefecture and year, so it enters the estimates only if longer-lived users from a prefecture travel to systematically different African destinations than its other users do.

⁵Weibo requires users to disclose the province and prefecture associated with their registration location, and only 9.2% of users in our sample report their registration region as ‘Others’.

The sample is defined by having posted from Africa at least once, so it allows no comparison with users who never left China or who traveled only to other world regions.⁶ The additional 3,089 accounts collected in 2022 were identified by searching Weibo for posts tagged with named locations in Africa, such as hotels, landmarks, businesses, or administrative places, during the 2018–2022 period. An account entered this second sample because at least one of its posts was associated with one of these African locations. We then retrieved the account’s complete posting history, including posts from before the search period. Unlike the original random draw of ten million users, this second sample is therefore conditional on having observed the user posting from Africa and should not be interpreted as a random sample of Weibo users.

The raw Weibo archive contains 76,309,599 posts from 93,971 users in 764 source partitions. Of these posts, 75,735,583 have a valid date, 7,796,578 contain usable coordinates, and 577,181 are assigned to an African destination after geographic harmonization. Among users with valid dates, the average interval between posts is 3.75 days; the average interval between geolocated posts is 17.33 days, and the average interval between posts observed in Africa is 24.07 days. These gaps show why we interpret the data as repeated signals of presence rather than continuous location histories.

Each spell is assigned to a destination defined by GADM administrative boundaries. We use ADM2 units wherever GADM defines that level and fall back to ADM1 in the six countries where it does not: Comoros, Cabo Verde, Libya, Lesotho, Mauritius, and Seychelles. Where raw ISO3 and administrative labels are missing in the six fallback countries, we assign the post to ADM1 by testing its coordinates against the corresponding GADM polygons.

The estimation panel is balanced by construction. It comprises all 6,475 GADM ADM2 units in the non-fallback countries together with all 95 ADM1 fallback units, giving 6,570 administrative units across 54 countries, each observed in every year from 2011 to 2023. Units in which no post is observed remain in the panel with zero outcomes. Over the full period only 1,227 of the 6,570 units record even one country spell, so 81.3 percent of the panel is never observed.

Observed travel is concentrated in a small number of destinations. We define a country spell as a distinct observed stay by a user in an African country, constructed from consecutive geolocated posts associated with that stay. Egypt, Morocco, and South Africa account for 25,152, 18,663, and 13,600 baseline country spells, respectively, or 42.6 percent of the total. The administrative units absorbing most of these spells rarely host an ongoing Chinese project. They are predominantly tourist destinations or, in the case of South Africa, locations with a large Chinese diaspora. In a comparison of project and non-project locations, such destinations would therefore make up a substantial part of the control group.

⁶We cannot calculate the exact sampling fraction because Weibo does not publish the total number of registered accounts from which our ten million-user sample was drawn. The widely cited figure of 510 million refers instead to monthly active users in September 2020 (Weibo, 2021) and therefore does not provide the appropriate denominator for the sampling rate.

Our principal outcomes are the number of distinct users and the number of new mobility spells observed in a destination-period. These intensive-margin counts capture how strongly an origin-destination link is used. We also measure the extensive margin with *Any user* and *Any new spell*, indicators for whether any presence or new spell is observed, and report the number of posts as a noisier measure that additionally reflects posting intensity.

4.2 Mobility spells

We define spells at the user-country level. For each user, posts are ordered chronologically and observations within the same African country are combined into one spell as long as no more than 30 days elapse between consecutive georeferenced observations associated with that stay. A new spell begins when the user is first observed in the country, returns after an observed stay elsewhere, or reappears following a gap of more than 30 days. We record the first and last observed dates of each spell as well as the number of days on which the user is observed. To locate a country spell within Africa, we assign it to the administrative unit in which the user is observed most frequently during that spell rather than assigning it to the country’s geographic centroid.

Defining spells at the country level is particularly important for our application. A definition based directly on movements across local administrative boundaries would potentially divide a single stay in an African country into several separate spells. For example, a person who temporarily leaves an administrative unit and subsequently returns would be recorded as starting another spell even if the movement formed part of the same journey. This would mechanically increase the number of spell starts and make their recorded timing depend on short-distance movements within the destination country. Such timing noise is especially problematic for our analysis because we relate the beginning of mobility episodes to narrowly defined monthly windows around labor unrest in China. The country-level definition instead treats within-country travel as part of the same international mobility episode and assigns one economically interpretable start date to that stay.

This procedure produces 134,711 country spells involving 69,532 users and 1,227 African destination units. The mean observed spell duration is 9.1 days, while the median is one day. Overall, 64.1 percent of spells last one observed day, 19.4 percent last more than one week, and 8.2 percent last more than four weeks. Egypt, Morocco, and South Africa are the most frequent destination countries. Because users may post infrequently or may not attach geolocatable information to every post, observed spell durations need not represent the full duration of a stay. We therefore interpret them as lower-bound measures of time spent at the destination.

The spell-based outcomes are an indicator of whether at least one new spell begins in a destination-period and the number of new spells beginning there. The former captures the extensive margin of newly observed mobility, while the latter measures the intensity with which new mobility episodes reach a destination. Throughout the paper, both outcomes are based exclusively on the country-level spell definition described above.

4.3 Origin profiles and exposure shares

User origins are obtained from the location information recorded in users’ Weibo profiles. Of the 93,254 users with a unique numeric identifier, 62,008 report a valid Chinese location and 61,093 can be assigned to one of 296 Chinese prefectures. We use these self-reported profile locations rather than inferring origins from users’ posting histories. Origin assignment is therefore constructed independently of the observed mobility outcomes in Africa.

For destination d and Chinese prefecture o , we calculate a link s_{do} from the users connecting that origin to that destination. The shares are normalized across African destinations within each Chinese origin, so $\sum_d s_{do} = 1$. They describe the African destination distribution of users from origin o in the full mobility sample. A destination receiving a larger fraction of an origin’s users receives more of that origin’s labor shock in the aggregate exposure.

Our standard shares use the full 2011–2019 link sample and are therefore not claimed to be predetermined. This is admissible under the shift-share interpretation when the identifying assumption concerns the shocks conditional on fixed effects rather than requiring the shares themselves to be randomly assigned (Goldsmith-Pinkham, Sorkin and Swift, 2020; Bluhm et al., 2025). As an additional safeguard, we construct leave-one-year-out shares: when estimating exposure in year t , we exclude year t from the sample used to form the link. This removes direct mechanical feedback from contemporaneous mobility into its own exposure measure, although it does not turn the shares into an exogenous treatment.

4.4 Labor shocks in China

Construction in China employs rural migrant workers through layered subcontracting chains that defer wage payments until the end of a project or year, an arrangement that generates recurrent wage-arrears disputes (Lee, 2007). We use the prefecture-year strike counts harmonized by Joris Mueller (2025), which combine China Strikes (Manfred Elfstrom, 2017) for 2003–2011 with the China Labor Bulletin strike map for 2012–2019 and cover 12,625 events. Wage arrears account for approximately three-quarters of recorded incidents since 2011, construction is consistently the largest sector, and the recorded events are small and localized (China Labour Bulletin, 2019; Elfstrom, 2021). The counts are compiled from worker and media reports, so they are a lower bound on true incidence, and we use them for relative variation across prefectures and years rather than as a measure of levels.

The baseline shock is the annual number of construction-sector strikes in prefecture o . Construction is the most directly observable sector for the implementation of infrastructure and other place-based development projects. A robustness measure uses all strike categories. The raw count is converted into an annual prefecture-level shock using the transparent transformations in the replication code; no intermediate canonical strike files are used. For the monthly timing test, we return to Mueller’s

cleaned event-level file and retain the exact date of each event. Aggregating these events reproduces the annual analysis file without a single city-year discrepancy: the 2014–2019 sample contains 4,062 construction strikes and 10,769 strikes across all categories.

4.5 Chinese development projects

Project locations, dates, donors, and sectors come from the Global Chinese Development Finance Dataset, Version 3.0 (Custer et al., 2023; Dreher et al., 2022). We spatially link geocoded projects to the harmonized African administrative units. A destination is active in year t if at least one project is between its recorded implementation start and completion years. For missing completion dates, the construction code applies the documented conservative rules rather than treating the project as indefinitely active. We reconstruct World Bank activity with exactly the same geographic assignment and timing rules for the donor-placebo exercise.

In the 2014-2019 estimation sample, 8.0 percent of administrative unit-years contain an active project and 6.7 percent contain a project active in both the preceding and current year. The Any-user outcome equals one in 26.1 percent of active-project unit-years and 7.7 percent of inactive-project unit-years. These raw differences are descriptive; the regressions absorb permanent destination differences and country-specific annual conditions.

For the sector analysis, we classify transport, energy, water and sanitation, communications, and industry/mining/construction as physical or labor-intensive. Health, education, government and civil society, social infrastructure, emergency response, food assistance, reconstruction, disaster preparedness, and population programs form a social/public group; remaining categories form an other productive/multisector group. To make “already under way” especially strict, this exercise requires the same project to be active in both $t - 1$ and t . The estimation sample contains 924 destination-years with an ongoing physical-sector project and 403 destination-years with only non-physical ongoing projects.

4.6 Platform coverage

Weibo coverage grows markedly over the sample period, making it important to distinguish geographic incidence from the number of sampled accounts. Our main country-by-year fixed effects absorb country-specific annual changes common to all destinations. We also construct two explicit coverage adjustments.

First, the coverage-standardized outcome converts the observed user count into the probability of at least one user at a common reference platform size:

$$\tilde{Y}_{dt} = 1 - \exp \left(-N_{2014:2019}^{\text{med}} \frac{\text{Users}_{dt}}{N_t} \right), \quad (7)$$

where N_t is the number of active sample users in year t and $N_{2014:2019}^{\text{med}} = 76,535$ is the median number of active sample users over 2014–2019. Second, we examine users per 10,000 active sample users. The first retains the extensive-margin interpretation; the second is a rate and can behave differently when user activity becomes geographically dispersed.

5 Empirical Strategy

Equation (8) estimates the reallocation parameter θ of Section 3 from annual origin-destination data. Equation (9) uses exact strike and spell dates. Section 5.3 tests whether θ varies with the operator’s recruitment base, the financing source, project completion, and the migrant network. Section 5.4 moves to the destination level, where origin totals cannot be held fixed. The project-start event study is in Appendix D.

5.1 The bilateral specification

Let L_{ot} denote the number of recorded strikes in prefecture o and year t , and let $O_{dt} = 1$ if destination d contains a project active in both $t - 1$ and t . We estimate

$$E[Y_{odt} \mid \cdot] = \exp \{ \alpha_{od} + \eta_{ot} + \mu_{dt} + \theta (L_{ot} \times O_{dt}) \} \quad (8)$$

by Poisson pseudo-maximum likelihood (PPML). O_{dt} is the binary counterpart of placement capacity K_{dt} in equation (5), and L_{ot} proxies the distress that site operators act on, scaled by the probability that an episode enters the record. The pair fixed effects α_{od} absorb the sunk cost F_{od} and the level of placement capacity along the corridor. The origin-by-year fixed effects η_{ot} absorb M_{ot} , the share of movers choosing Africa, and the strike count itself. The destination-by-year fixed effects μ_{dt} absorb destination payoffs, project status, and local platform coverage. θ is the differential in mobility from o toward its ongoing-project destinations, relative to its other African destinations, per additional strike in o . Standard errors are clustered by origin prefecture.

Because ω_{od} and $\bar{A}_{od}$ vary across pairs, $\hat{\theta}$ averages over corridors with different recruitment links and network thickness, weighted by expected counts. High-volume tourist corridors therefore receive substantial weight despite rarely containing projects, which attenuates the estimate. Section 5.3 directly examines heterogeneity in the operator’s recruitment links.

We also estimate a lead version, with mobility and project activity measured in $t - 1$, strikes in t ,

and the fixed effects aligned with the outcome year. An operator that recruits in o adjusts staffing when accumulated grievance crosses τ_F , before workers strike at τ_S , so mobility toward linked sites should already be higher in the year before a strike is recorded. A positive lead is also what one would see if the destination composition of some prefectures was shifting toward project locations before their strike counts rose. The annual data cannot separate the two. The monthly estimates in Section 5.2 and the split by the implementing firm’s home prefecture in Section 5.3 address this.

The pair-year panel contains every origin-destination pair observed at least once during 2011–2019, crossed with the six analysis years 2014–2019. Pairs never observed in that window are not in the panel, so θ is estimated from corridors that exist. The underlying pair-year panel contains 106,152 observations, corresponding to 17,692 bilateral pairs, 295 Chinese origins, and 1,355 African destinations. Binary outcomes indicate whether a pair contains any user or any newly starting country spell; count outcomes are users, posts, and new spells. PPML accommodates zero outcomes, but observations that are perfectly separated do not contribute to estimation. The effective contemporaneous samples therefore contain 86,012 observations for the user and post outcomes and 58,363 observations for the spell outcomes. In the annual lead specifications, the corresponding sample sizes are 76,098 and 51,394 observations. We report $100[\exp(\hat{\theta}) - 1]$ throughout, the percentage difference in a pair’s expected outcome in ongoing-project destinations relative to other destinations per additional strike.

The origin-by-year fixed effects absorb changes in total observed mobility from o , so equation (8) identifies changes in the allocation of that mobility across African destinations. To examine the aggregate margin separately, we construct an origin-prefecture-by-year panel and estimate PPML models of unique users, posts, spells, and the number of African destinations reached on L_{ot} , with origin and year fixed effects. These models test whether strikes are associated with changes in total observed mobility from an origin to Africa rather than with its allocation across destinations.

5.2 Timing within the year

Annual data cannot determine whether a spell begins before or after a strike in the same year. We therefore construct an origin-prefecture-by-African-destination-by-month panel using exact spell-start and strike dates. For outcome month m , we estimate

$$E[Y_{odm} \mid \cdot] = \exp \left\{ \alpha_{od} + \eta_{om} + \mu_{dm} + \sum_{k \in \mathcal{K}} \theta_k L_{o, W_k(m)} \times O_{d, t(m)} \right\}, \quad (9)$$

where $O_{d, t(m)}$ equals one only if a project is active in both the preceding and current calendar year. The five jointly included strike windows are months $m + 1$ through $m + 3$, month m , months $m - 1$ through $m - 3$, months $m - 4$ through $m - 6$, and months $m - 7$ through $m - 12$. The window coefficients show when, relative to the recorded strike, mobility toward ongoing-project destinations changes. In the first window mobility precedes the strike by one to three months. Under the

framework, an operator that recruits in o adjusts once τ_F is crossed, which places its response in this window, whereas operators without arrangements in o draw on the prefecture only after the strike. Ordering within the current month is ambiguous. Pair, origin-by-month, and destination-by-month fixed effects are included, and standard errors are clustered by origin prefecture.

The monthly outcomes are the number of spell-start users, the number of spell starts, and an indicator for any spell-start user. The panel covers January 2014 through September 2019; the final three months are excluded because the one-to-three-month pre-strike window cannot be observed for them. We estimate equation (9) first for construction strikes and then for all strike categories.

We also split spells by whether the user is observed in one or in several harmonized African administrative units. Tourists and traders are in the sample and do not respond to project capacity. PPML identifies a proportional effect, so their share in a pair scales the estimate toward zero. The estimate in the single-unit subsample is closer to the effect among responsive users to the extent that tourists are underrepresented there; the split does not classify anyone as a worker.

5.3 Tests of the framework’s predictions

The framework predicts that θ is larger where the operator recruits in o , that the response leads the strike where such a link exists and lags it where it does not, that the differential disappears when the project ends, and that it is larger where the migrant network is thin. Each test below splits the strike-by-project interaction of equations (8) and (9) accordingly.

Operator recruitment base. We use the contract-winner data assembled by Mueller (2025) to link Chinese-financed projects to their implementing firms and the firms’ home prefectures. We define a project location as linked to origin o if at least one ongoing project is implemented by a firm based in o . We then split the strike-by-project interaction in equations (8) and (9) into linked and unlinked components and test their difference using the estimated covariance matrix. The sample remains the full origin-destination panel, including non-project locations. Our preferred definition matches firms and origins at the prefecture level; because recruitment and subcontracting networks may extend beyond a firm’s registered prefecture, we also report province-level matches. All projects used in the split were already active before the relevant strike, separating this test from the allocation of new contracts following unrest.

The framework predicts a larger pre-strike response at project locations whose implementing firm is based in the affected prefecture or province. Because contractor coverage is incomplete, an unlinked observation can indicate either a firm from another region or an unidentified implementer. We therefore also report a sensitivity specification excluding project-years for which no contractor origin is observed.

Donor comparison. We repeat the estimation with World Bank project activity in place of Chinese project activity, holding strike counts, outcomes, sample, and fixed effects unchanged, and also enter both interactions jointly. We do this in the annual model, the monthly model, and the destination-level model of Section 5.4. Chinese firms build a substantial share of multilateral civil works in Africa (Foster et al., 2009), so a World Bank site may be staffed by Chinese personnel; the comparison is not one of contractor nationality. Contracts allocated through MOFCOM go disproportionately to provincially anchored firms, while competitively procured works go more often to central state-owned enterprises recruiting nationally, so a recruitment link to the shocked prefecture is more likely at a Chinese-financed site. The framework then predicts a lead at Chinese-financed sites and a lag at World Bank sites. The two project indicators overlap and World Bank locations are more numerous, so the joint models separate the two types of operator by timing rather than isolating the financing source.

Project completion. The strike differential should be strongest while a project provides active implementation capacity. We therefore replace the binary ongoing-project indicator in equation (9) with two mutually exclusive indicators: a project currently under implementation and a project completed within the preceding k years. Destinations with no previously active project or with projects that have not yet started form the omitted category. Each of the five strike windows is interacted jointly with both project states. A completed state begins in the calendar year after the final active year; where projects overlap, an ongoing project takes precedence.

The baseline defines recently completed as the two years following completion and uses the actual completion year where available and the planned completion year otherwise. We report one- and three-year windows and an actual-completion-only classification as sensitivity checks. In addition to the state-specific coefficients, we test directly whether the ongoing and completed coefficients are equal. The estimation sample, PPML estimator, fixed effects, and origin-prefecture clustering are unchanged from equation (9). Comparable actual and planned completion dates are unavailable for World Bank projects, so this exercise is restricted to Chinese projects.

Network thickness. The framework predicts a larger θ where the migrant network from o at d is thin. A network measure built from the mobility data would produce mean reversion in the direction of the prediction, so the measure has to come from elsewhere. Province-country twinning arrangements are dated and assigned outside the Weibo record, and we interact the strike-by-project term with an indicator for a twinning arrangement between o 's province and d 's country. We also compute origin-specific destination shares from 2011–2013 alone and interact them with strikes in the origin-destination panel, with the same fixed effects, which asks whether a shock reinforces the routes that were strongest before the estimation window. Platform coverage was thinnest in those years, weak links are measured with error, and the resulting mean reversion runs in the direction of the prediction, so we do not rely on this exercise.

We also split ongoing projects by sector, since the rotation margin should be more elastic at labor-intensive sites. Sector is not randomly assigned and the framework does not model it. Appendix C.1 reports the split as exploratory evidence.

5.4 Destination-level exposure

The bilateral model holds each origin's total fixed. A destination-level model instead asks how much Chinese presence an African district records when the prefectures connected to it come under pressure, reallocation and volume combined. Let s_{do} denote origin o 's full-sample destination share allocated to African destination d . Destination-year exposure is

$$Z_{dt} = \sum_o s_{do} L_{ot}. \quad (10)$$

Exposure is high when strikes occur in prefectures that send a large share of their observed African mobility to d . It varies across destinations through the shares and over time through the strike counts. In levels, the destination-level model is

$$Y_{dct} = \alpha_d + \psi_{ct} + \beta Z_{dt} + \delta O_{dt} + \gamma (Z_{dt} \times O_{dt}) + u_{dct}, \quad (11)$$

with destination and country-by-year fixed effects. Because Z_{dt} aggregates across origins, γ combines the reallocation θ with any change in the total number of people the linked prefectures send abroad. The origin-by-year fixed effects in equation (8) remove the second component; the destination and country-by-year fixed effects here cannot, since both components vary at the destination-year level.

The shares are constructed from the same mobility data as the outcome. We therefore also report leave-one-year-out shares that exclude year t from the sample used to form the link. This removes the mechanical link between current mobility and its own exposure measure. The shares remain endogenous.

The levels model does not date the response. In the main text we therefore report local projections,

$$Y_{dc,t+h} - Y_{dc,t-1} = \alpha_d^h + \psi_{ct}^h + \beta_h Z_{dt} + \delta_h O_{dt} + \gamma_h (Z_{dt} \times O_{dt}) + u_{dct}^h, \quad (12)$$

for horizons $h = 0, 1, 2, 3$, and a lead with $Y_{dc,t-1} - Y_{dc,t-2}$ as the dependent variable, labelled $h = -1$. γ_h is the additional change per unit of exposure in locations where a project was already under way. Exposure is continuous, project status changes, and the shock recurs, so this is not a binary difference-in-differences design, and the local-projection form does not impose a single persistent response. $O_{dt} = A_{d,t-1} A_{dt}$ is defined as in the annual model; a project awarded in t has no site in $t - 1$ and belongs to the contract margin that Mueller (2025) documents. The appendix reports the broader A_{dt} definition, which admits projects starting in t , together with the levels estimates, sample-start sensitivity, and a split of destination-years by prior observed presence (Appendix I).

Standard errors are clustered by African destination country, which allows dependence among administrative units within a country; destination-unit and two-way destination-country and destination-unit clustering are reported as sensitivity checks. Because Z_{dt} is a shift-share, we also decompose it into its origin contributions and report an origin-shock score calculation (Adão, Kolesár and Morales, 2019; Borusyak, Hull and Jaravel, 2022). This concern does not arise in the bilateral models, where the strike count enters directly and the origin is the cluster. Country clustering and the shock-side calculation address different dependence structures; the main tables use the former, which is more conservative.

6 Results

6.1 Annual reallocation toward projects already underway

Figure 1 reports the direct allocation estimates from equation (8) along with the annual lead estimates. Appendix Table 2 reports both series numerically. Appendix Table 1 compares the contemporaneous estimates with the broader active-in- t definition. A positive coefficient means that mobility from a higher-strike origin is relatively more oriented toward destinations where projects were already underway.

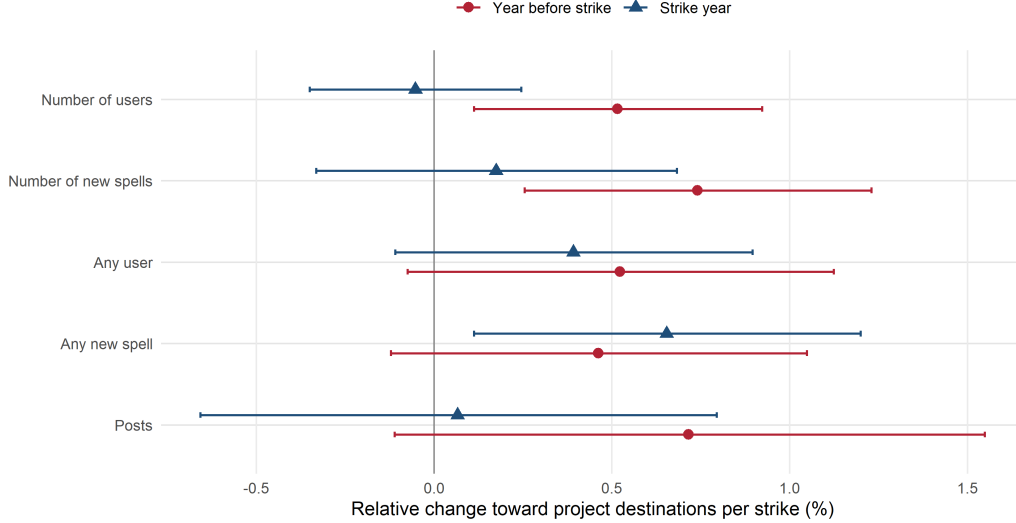


Figure 1: Origin-destination reallocation toward project locations

Notes: Points show $100[\exp(\hat{\theta}) - 1]$. The strike-year series uses mobility and strikes in t and requires a project to be active in both $t - 1$ and t . The lead series uses mobility and project activity in $t - 1$ and strikes in t , without conditioning on project survival into t . Bars represent 95 percent confidence intervals based on standard errors clustered by Chinese-origin prefecture.

The clearest annual result appears in the counts before the strike is recorded. In $t - 1$, one additional construction strike subsequently observed in t is associated with 0.52 percent more users in project

destinations relative to other destinations reached from the same prefecture (standard error 0.21, $p = 0.012$) and 0.74 percent more spells (standard error 0.25, $p = 0.003$). Requiring the same project to be active in both $t - 2$ and $t - 1$ produces similar estimates of 0.49 percent for users ($p = 0.038$) and 0.70 percent for spells ($p = 0.022$). The corresponding binary leads are positive but imprecise, at 0.52 percent for Any user ($p = 0.086$) and 0.46 percent for Any new spell ($p = 0.121$).

In the strike year, the pattern shifts toward the binary outcomes. One additional construction strike is associated with a 0.65 percent higher expected probability of recording a new mobility spell in ongoing-project destinations relative to other destinations (standard error 0.28, $p = 0.018$). The Any-user coefficient is positive but imprecise at 0.39 percent. The user and spell-count coefficients are close to zero. Mobility does not jump when the strike is registered. It reorients toward existing project destinations in the year before; in the strike year itself, the change is that additional pairs record mobility at all. Consistent with Section 3, we read the count and binary outcomes as closely related functions of thin data rather than as evidence on distinct margins. Origin-by-year fixed effects imply that both patterns are changes in destination composition rather than additional total mobility.

The strike-year result is donor-specific. In the joint model, the Chinese-project Any-new-spell differential remains 0.65 percent in the strike year (standard error 0.29, $p = 0.024$), whereas the World Bank differential is 0.01 percent (standard error 0.40, $p = 0.975$). The annual lead does not separate the donors. Neither Any-new-spell coefficient is significant, at 0.37 percent for Chinese projects (standard error 0.31, $p = 0.243$) and 0.51 percent for World Bank projects (standard error 0.49, $p = 0.296$). Appendix Table 7 reports the comparison. Annual aggregation cannot distinguish a lead of a few months from a contemporaneous response. We therefore estimate the same interaction at monthly frequency.

6.2 Within-year timing

Figure 2 examines whether the annual reallocation can be located more precisely within the year. It plots equation (9) with the number of spell-start users and the number of spell starts first, followed by the Any-user outcome.

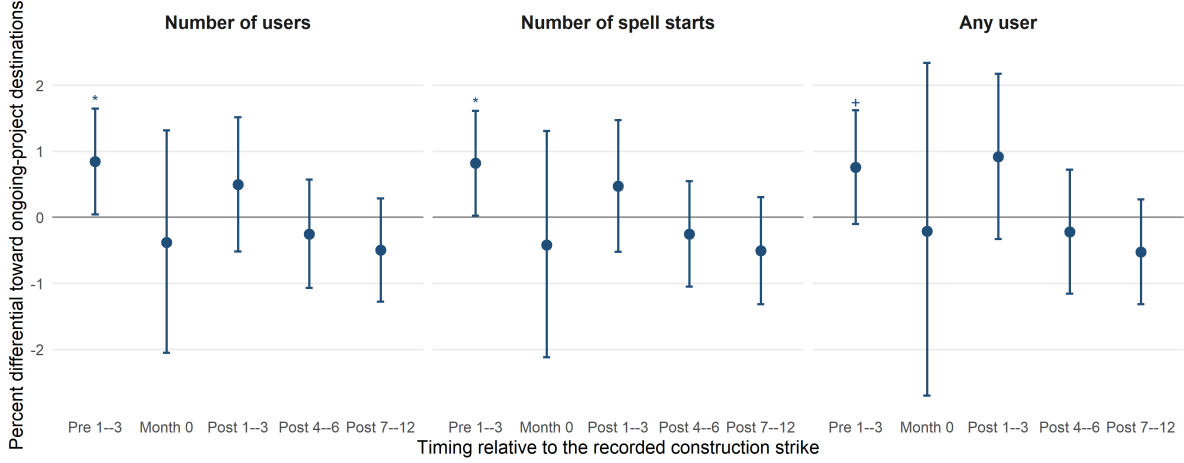


Figure 2: Monthly timing of construction strikes and mobility toward ongoing-project destinations
Notes: Points report the percentage differential in the expected outcome in destinations with a project active in both $t - 1$ and t , per additional construction strike in the indicated window. All five windows enter jointly. In the leftmost coefficient, mobility precedes the strike by one to three months; under the framework of Section 3, this window captures operators that recruit in the shocked prefecture, whereas post-strike windows capture sites drawing on the prefecture only once a strike has occurred. Models include origin-destination-pair, origin-month, and destination-month fixed effects. Bars are 95 percent confidence intervals based on standard errors clustered by Chinese origin prefecture. Stars denote $p < 0.05$; plus signs denote $p < 0.10$.

The monthly estimates place the statistically detectable construction-strike pattern before the recorded event. One to three months beforehand, the relative differential toward ongoing-project destinations is 0.85 percent for the number of spell-start users (standard error 0.41, $p = 0.038$) and 0.82 percent for the number of spell starts (standard error 0.40, $p = 0.043$). The Any-user estimate is 0.76 percent and only marginally significant (standard error 0.44, $p = 0.084$). None of the same-month or post-strike construction coefficients through twelve months is significant. The reorientation toward ongoing projects precedes the recorded strike. Appendix Table 4 reports the complete set of monthly estimates. The estimates give no sign that authorities or firms send people to Chinese-financed sites after observing one. Appendix Table 6 examines whether the monthly response differs between spells in which a user is observed in only one harmonized African administrative unit and spells spanning several units. In the one-to-three-month pre-strike window, the estimated response for single-unit spells is 3.32 percentage points larger for the number of spell-start users and 3.27 percentage points larger for the number of spell starts than the corresponding response for multi-unit spells. Both differences are statistically significant ($p = 0.039$ in each case). For the probability of observing any spell start, the estimated difference between the two groups is 1.86 percentage points and is not statistically significant ($p = 0.328$). Thus, the two types of spells differ significantly in their count responses, but not in the binary outcome. Because these differences are not stable across later windows and we do not observe occupation or reason for travel, we treat the exercise as a behavioral diagnostic rather than as evidence that the responding users are project workers.

The donor comparison at monthly frequency separates the two project types by timing. With the

Chinese- and World-Bank-project interactions entering jointly, the Chinese pre-strike estimates are 0.88 percent for any spell start ($p = 0.088$), 0.87 percent for users ($p = 0.045$), and 0.84 percent for spell starts ($p = 0.049$). The corresponding World Bank estimates are -0.38, -0.05, and -0.04 percent and are all statistically insignificant. In the one-to-three-month post-strike window, by contrast, the Chinese coefficients are insignificant while World Bank locations show 1.06 percent more users ($p = 0.013$) and 1.03 percent more spell starts ($p = 0.017$). The same World Bank post-strike pattern appears when all strike categories are used. Appendix Figure 4 and Table 5 report all joint estimates.

The lead-lag pattern matches the timing structure in Section 3. The lead appears at Chinese-financed sites, where operators with a recruitment link to the shocked prefecture are more likely to be present. The lag appears at World Bank sites, which Chinese contractors frequently implement (Foster et al., 2009) and where placement capacity exists without such a link. A lead at World Bank sites would have contradicted the mechanism; the estimates show none. The comparison is indirect. It sorts sites by who financed the project, not by where the operator recruits, and we do not observe the recruitment base in these models. In Section 5.3 we test the mechanism more directly by splitting project locations according to the implementing firm’s home prefecture.

6.3 Operator recruitment base

The operator-origin split provides partial support for the recruitment mechanism. In the year before a recorded strike, one additional future strike is associated with 0.94 percent more users at project locations whose implementer is based in the affected prefecture ($p = 0.013$), compared with 0.46 percent at project locations without such a match ($p = 0.021$). Although the estimate is larger at linked locations, the difference between the two coefficients is not statistically significant ($p = 0.173$).

The pattern is stronger for mobility spells. The estimated increase is 1.44 percent at linked project locations ($p < 0.001$), compared with 0.62 percent at unlinked project locations ($p = 0.010$). The difference between these estimates is marginally significant ($p = 0.052$). We find no corresponding positive result on user or spell counts in the strike year.

The results become clearer when recruitment links are defined at the province rather than the prefecture level. The pre-strike response at provincially linked locations exceeds that at other project locations by 0.86 percent for users ($p = 0.026$) and by 1.07 percent for spells ($p = 0.005$). One reading is that the relevant recruitment, administrative, or subcontracting networks extend beyond the implementing firm’s registered prefecture.

The monthly estimates display the same ordering but are less precise. One to three months before a recorded strike, user counts increase by 1.69 percent at same-prefecture linked project locations and by 0.71 percent at project locations without such a match. The corresponding estimates for spells are 1.60 and 0.69 percent. The differences between the linked and unlinked coefficients are not statistically significant for either users ($p = 0.302$) or spells ($p = 0.320$). The remaining monthly

windows do not reveal a consistent separation between the two groups.

The results suggest that pre-strike mobility responds more strongly where the implementing firm is regionally connected to the affected Chinese origin, particularly for spells and when connections are defined at the province level. They do not show that the response is confined to project locations with an exact same-prefecture implementer. The exact-match comparison also rests on thin coverage. Only 39 percent of ongoing Chinese project-years have an observed geographically assigned implementer. The exact-match linked group covers 25 origin prefectures.

6.4 Project completion

We next ask whether the one-to-three-month pre-strike differential depends on the project still being under implementation. We divide project destinations into those with an ongoing project and those at which a project was completed within the preceding two years and no other Chinese project remains ongoing, and split the pre-strike interaction by these two project states. Both interactions enter the same PPML regression, with origin–destination-pair, origin-by-month, and destination-by-month fixed effects.

At ongoing-project destinations, the pre-strike differential amounts to 0.86 percent for users, 0.82 percent for new spells, and 0.75 percent for Any user. At recently completed destinations, the corresponding estimates are negative and statistically indistinguishable from zero. The differential is detectable while projects are under implementation but not after completion.

The comparison is suggestive rather than decisive. Under the baseline two-year completion window, direct tests cannot reject equality between the ongoing and completed coefficients. When the completed category is extended to three years after completion, the difference becomes statistically significant for users ($p = 0.038$) and new spells ($p = 0.031$), and marginally significant for Any user ($p = 0.061$). The results are consistent with implementation-specific capacity, but the formal evidence depends on where the post-completion period ends.

6.5 Destination-level dynamics

Figure 3 reports the destination-level estimates from equation (12) as a robustness exercise. Unlike the bilateral model, it aggregates origin strike counts using historical destination shares, so it measures the composite of reallocation and volume described in Section 5.4. At $h = -1$, the outcome is the change from $t - 2$ to $t - 1$; at $h = 0, \dots, 3$, it is the change from $t - 1$ to $t + h$. The project indicator in the plotted Any-user specification requires activity in both $t - 1$ and t . We retain this model because it asks whether the bilateral allocation result is visible in aggregate destination outcomes. It is not the paper’s primary design.

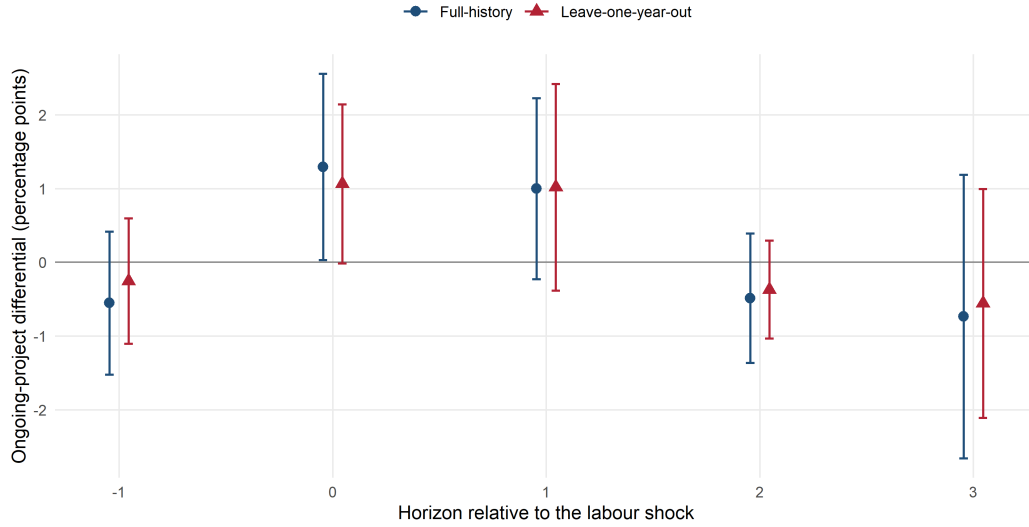


Figure 3: Dynamic Any-user response in ongoing-project locations

Notes: Points show $\hat{\gamma}_h$, the additional response in ongoing-project locations. At $h = -1$, the dependent variable is $Y_{t-1} - Y_{t-2}$; at $h = 0, \dots, 3$, it is $Y_{t+h} - Y_{t-1}$. Bars are 95 percent confidence intervals based on African-country clustered standard errors and finite-cluster t critical values.

The destination-level Any-user result is short-lived. At $h = -1$, the ongoing-project differential is -0.55 percentage points with full-history shares and -0.25 percentage points with leave-one-year-out shares; neither is significant. At $h = 0$, the corresponding differentials are 1.29 percentage points ($p = 0.045$) and 1.06 percentage points ($p = 0.053$). The estimates remain near one percentage point but become imprecise at $h = 1$. They are negative and insignificant at $h = 2$ and $h = 3$. Appendix Table 3 reports the corresponding numerical estimates.

The count dynamics do not cleanly replicate the annual and monthly bilateral results. In the broader active-in- t specification, the immediate total responses at project locations are -10.26 users and -11.14 new spells per unit of exposure, both of which are statistically significant. At $h = 1$, however, the active-project differentials turn strongly positive, at 32.84 users and 39.16 spells. They remain positive at $h = 2$ and $h = 3$. Because mobility falls outside active project locations over these horizons, the resulting total effects within project locations are small and statistically indistinguishable from zero. The dynamic counts indicate relative cushioning or reallocation toward project locations, not an absolute increase in mobility. Appendix Tables 15, 16, and 21 report the corresponding dynamic specifications and immediate count decomposition; the remaining horizons are discussed as sensitivity evidence.

The destination-level pattern is not specific to Chinese-financed projects. Replacing Chinese projects with World Bank projects yields a 1.51-percentage-point Any-user differential at $h = 0$ and positive differentials through $h = 3$. When both donor interactions enter together, the World Bank coefficients are positive and statistically significant or marginally significant throughout, whereas none of the positive Chinese coefficients are significant. Appendix Figure 5 and Table 8 report

the comparison. The destination-level coefficient has the composition described in Section 5.4. The model absorbs no origin-specific information, so it cannot capture the timing asymmetry that distinguishes the two types of operators. World Bank locations are also more widespread; the two project indicators overlap. Chinese contractors can also implement World Bank-financed civil works, so a Chinese-personnel channel is not excluded. We use this model as secondary evidence that project locations cushion or redirect mobility, not as China-specific identification.

6.6 Aggregate mobility

The reallocation results do not imply that labor unrest increases total mobility from China to Africa. The framework is silent on the volume response and gives reasons to expect it to be small. Appendix Table 22 aggregates outcomes across all African destinations within each Chinese origin-year. One additional construction strike is associated with 0.28 percent more users and 0.37 percent more spells. Neither estimate is statistically distinguishable from zero. Both take the same sign as the bilateral estimates and are of a similar order of magnitude, so the origin totals are too imprecise to rule out either a small volume response or none at all. Neither the number of destinations reached nor the number of destinations observed for the first time increases. Posts rise in one construction-strike specification, but the estimate disappears with all strike categories and becomes only marginal after normalization by total archive posts. We find a change in relative destination choice without a robust change in aggregate observed mobility.

6.7 Robustness and Supplementary Evidence

The donor comparison gives mixed results. At monthly frequency, the Chinese pre-strike user and spell-count coefficients remain positive and the World Bank pre-strike coefficients are near zero, but World Bank locations exhibit positive count responses one to three months after the strike. In the annual strike-year origin-destination model, the Any-new-spell result is present at ongoing Chinese projects but absent at World Bank projects when both interactions enter jointly. The destination-level dynamic model also generates positive World Bank coefficients. Because Chinese contractors frequently implement multilateral civil works, the World Bank comparison cannot rule out a Chinese-personnel response; it can only show whether results are confined to Chinese-financed locations.

Sector heterogeneity is consistent with, but does not establish, a personnel-demand mechanism. In the broad joint model, one additional construction strike is associated with 0.89 percent more expected Any-new-spell incidence and 0.71 percent more Any-user incidence at destinations with an ongoing physical or labor-intensive project. The corresponding estimates at destinations containing only non-physical projects are 0.30 and -0.06 percent. The formal physical-minus-non-physical difference is significant for Any user (0.77 percent, $p = 0.022$) but not for Any new spell (0.58 percent, $p = 0.139$). The detailed physical-only versus social/public-only Any-user contrast is 0.98

percent and marginally significant ($p = 0.066$). Restricting the sample to locations with ongoing projects produces smaller and insignificant sector differences. Appendix Figure 6 and Table 9 show the complete extensive-margin comparison. We treat it as exploratory because the sector split was not randomized, multiple outcomes are examined, and some smaller sector cells are sparse.

The remaining appendix results are informative but do not define the ongoing project estimand as cleanly. Replacing ongoing status with the broader indicator for any project active in t gives nearly the same origin-destination spell estimate, 0.62 percent, and a somewhat larger immediate destination-level Any-user differential of 1.68 percentage points (Appendix Tables 1, 15, and 16). The main pattern does not depend on the precise project-timing cutoff. Because this broader definition includes projects that begin in the shock year, however, it is less well matched to the existing-capacity mechanism.

Static destination-level estimates point in the same direction. Under the ongoing-project definition, the full-history differential is 0.89 percentage points ($p = 0.062$) and the leave-one-year-out differential is 0.59 percentage points ($p = 0.047$); Appendix Tables 17 and 18 report the corresponding static specifications. We place these level regressions in the appendix because their timing is less transparent than the change specification. The sample-start exercise is also in the appendix (Appendix Figure 9 and Table 19). Its non-monotonic precision is a further reason not to let the static model carry the main claim.

Broadening the shock to all strike categories preserves a positive annual extensive-margin association in the broader active-in- t models (Appendix Table 11). The annual spatial relationship does not appear to be purely a construction-sector classification artifact. The intensive-margin evidence is more timing-specific. User and spell counts are positive and significant in the annual and monthly pre-strike origin-destination models, close to zero contemporaneously, and unstable in the aggregated dynamic models.

Coverage standardization, alternative destination-side clustering, shock-side inference, exposure-tail checks, and leave-one-country and leave-one-origin exercises preserve a positive project-location differential in the specifications for which they were estimated (Appendix Table 20, Appendix Tables 13 and 14, and Appendix Figure 8 and Table 12). These checks are reported in the appendix because most use the broader active-in- t definition. They support the general spatial result but cannot substitute for direct inference on individual employment or establish that a recorded strike causes mobility.

7 Interpretation and Limitations

Project placement does not need to be random for θ to be interpretable, because in our interaction we do not compare project locations with non-project locations. The pair fixed effects hold the corridor fixed, the origin-by-year fixed effects hold the prefecture’s total observed mobility fixed,

and the destination-by-year fixed effects hold conditions at the destination fixed. The remaining variation is whether a corridor carries relatively more mobility toward a destination with an active project in years when its origin records more strikes. Anything that varies at the level of the interaction itself could still bias the estimate, such as a destination shock that hits project locations and is correlated with conditions in the origins linked to them.

Had a recorded strike itself set deployment in motion, mobility toward project sites would subsequently have increased. Instead, the differential appears one to three months before the strike, and no comparable lagged effect follows. An operator with recruitment arrangements in the prefecture can be expected to produce exactly this timing. It adjusts staffing once distress has sufficiently lowered workers' opportunity costs (e.g. visible in the recruitment wage) to cover the cost of a lumpy spending on increased staffing, which happens before workers have accumulated enough grievance to strike.

The same ordering would be produced by prefecture-level trends. Suppose construction strikes proxy for conditions that were already tilting a prefecture's destination mix toward project locations. The monthly coefficients would then look as documented in our results. In this case, there is the risk that our framework provides an incorrect explanation for the patterns in the data. However, two results speak against that incorrect interpretation being a problem in our case. We document that the lead is larger where the implementing firm is based in the affected prefecture or province. A trend common to the whole prefecture should not lead to sorting based on where the implementer comes from. In the destination-level part of the model, the differential at $h = -1$ is negative and insignificant under both share constructions, so project locations were not drifting upward in the year before exposure rose. Neither result settles the question because a trend concentrated in the corridors connecting a prefecture to firms in its own region cannot be ruled out.

World Bank locations show the reverse ordering, with pre-strike estimates close to zero. In the one to three months after a strike, users rise by 1.1 percent and spell starts by 1.0 percent. In the model, a site whose operator has no arrangements in a prefecture can hire from it only after a strike has made a pool of workers visible, so the two-threshold structure implies a lag at such sites. A lead at World Bank locations would have contradicted the mechanism. A post-strike rise there has other possible sources as well. Large project sites share subcontracting chains, business networks, transport links, and accommodation, any of which could draw mobility once a prefecture's workers become available, whether or not a Chinese contractor is involved. The comparison sorts sites by who financed them, and financing is only a noisy signal of where the operator recruits. The operator split gets closer to recruitment itself. In the year before a recorded strike, one additional strike is associated with 1.4 percent more spells at project locations whose implementer is based in the affected prefecture, against 0.6 percent at other project locations. The difference is marginally significant ($p = 0.052$). For users the ordering is the same and the difference is not significant ($p = 0.17$). Matching at the province level separates the groups more clearly, which suggests that the recruitment and subcontracting networks a firm draws on reach beyond its registered prefecture.

The exact match is thin. A geographically assigned implementer is observed for 39 percent of ongoing Chinese project-years and the linked group covers 25 origin prefectures, so an unlinked project may belong to a firm from elsewhere or to one we cannot place.

After a project ends, there is no longer a detectable pre-strike differential. Under the two-year definition of recent completion, the estimates at completed destinations are negative and statistically imprecise. Equality with the coefficients for ongoing projects cannot be rejected. With a three-year window the difference becomes significant for users and new spells. The port, mine, or city that made a location attractive is still there after completion, so a differential that persisted would have pointed to the place rather than to the project. The estimates point toward the project, with the caveat that the strength of that tentative evidence depends on where the post-completion window is cut.

None of this identifies the observed users as actual project workers. A Weibo profile gives a prefecture of registration and a geolocated post gives presence at an African destination. Occupation, employer, and reason for travel are typically not observed. The mechanism we have in mind runs through the pool of workers available to site operators and the terms on which they are recruited, which is why the estimates can be read without classifying anyone. They are changes in observed mobility intensity and should not be interpreted as a precise count of workers or migrants.

A further limitation is related to the fact that Weibo adoption grew over the sample period and only a minority of posts carry a location. The destination-by-year and country-by-year fixed effects, the 2014 sample start, the alternative sample windows, and the coverage-standardized outcomes make it unlikely that platform growth alone produces the differential. What they cannot rule out is a change in who, within a prefecture, posts from Africa. Our outcome is observed mobility on Weibo not total Chinese mobility to Africa.

Our destination-level exposure design adds an assumption that the bilateral models do not need. Historical shares reflect earlier migration and economic ties and can be related to underlying destination characteristics. That is valid when identification is assigned to the shocks rather than to the shares (Goldsmith-Pinkham, Sorkin and Swift, 2020), in which case the requirement is that, conditional on destination and country-by-year fixed effects, changes in origin-prefecture strike counts are not differentially related to unobserved destination shocks through those historical links. Leave-one-year-out shares remove the mechanical feedback from contemporaneous mobility into the exposure measure.

The destination-level estimates do not hold origin totals fixed and therefore combine reallocation with changes in volume; we use them only to check the aggregate pattern. The World Bank comparison is less direct because the financing source is an imperfect proxy for where the operator recruits. The completion results indicate that the differential depends on whether the project is active rather than on location. Our main evidence that active sites redirect mobility from prefectures under distress is therefore the monthly origin-destination estimates and the split by implementer origin, with the

annual estimates showing the same lead. A corridor-specific trend tied to the expansion abroad of firms based in the prefecture remains the reading these results do not exclude.

8 Conclusion

Chinese-financed development projects shape more than the allocation of capital across Africa. While they are under implementation, they also create temporary connections between particular African locations and firms, workers, and labor markets in China. This paper studies whether these connections affect where Chinese mobility appears within Africa when economic conditions deteriorate at home.

We combine individual mobility traces from Weibo with geocoded development projects across Africa and construction strikes in China. From 76.3 million posts by 93,971 users, we construct observed stays in African countries and link users to their Chinese prefecture of origin. Our main analysis covers 17,692 mobility corridors connecting 295 Chinese prefectures to 1,355 African destinations between 2014 and 2019. By comparing, within the same Chinese prefecture and period, mobility toward African locations with an ongoing Chinese-financed project to mobility toward other African destinations, we identify changes in the geography of Africa-bound mobility while holding total observed mobility from each origin fixed.

The main result is a reallocation of mobility across African destinations rather than a measurable increase in its total volume. In the annual data, one additional construction strike recorded in year t is associated with 0.52 percent more users and 0.74 percent more new mobility spells at ongoing-project destinations in $t - 1$, relative to other African destinations reached from the same prefecture. At monthly frequency, the timing is sharper: one to three months before a strike is recorded, ongoing-project destinations receive 0.85 percent more users and 0.82 percent more new spell starts per additional strike. Both estimates are statistically significant, while no comparable increase appears in the strike month or during the following twelve months. The effect is therefore not that deteriorating conditions in China generate substantially more mobility to Africa, but that they change which African locations receive that mobility.

The timing is consistent with construction strikes marking the visible culmination of labor-market deterioration rather than its onset. Firms connected to particular Chinese prefectures through recruitment or subcontracting relationships may observe worsening conditions before they culminate in collective action. The implementer results support this interpretation. In the year before a recorded strike, the increase in new mobility spells is 1.44 percent at African project locations whose implementing firm is based in the affected prefecture, compared with 0.62 percent at other project locations; the difference is marginally significant at $p = 0.052$. At the province level, the response at linked locations exceeds that elsewhere by 0.86 percent for users and 1.07 percent for spells. These patterns are consistent with project implementation linking specific African destinations to particular regional recruitment and subcontracting networks in China.

The comparison with World Bank-financed projects provides a second indication that these organizational links matter. When Chinese- and World-Bank-project interactions enter jointly, Chinese-financed sites show pre-strike increases of 0.87 percent in users and 0.84 percent in spell starts. At World Bank sites, the corresponding pre-strike estimates are close to zero. Instead, one to three months after the strike, World Bank locations record 1.06 percent more users and 1.03 percent more spell starts, while the Chinese-project coefficients are no longer statistically distinguishable from zero. Financing source does not identify firms' recruitment networks directly, and Chinese firms can also implement World Bank projects. The comparison is therefore indirect, but its timing is consistent with differences in how strongly particular African project sites are connected to particular Chinese origins.

The response also appears to depend on the project remaining active. At ongoing Chinese-project destinations, the one-to-three-month pre-strike differential is 0.86 percent for users and 0.82 percent for new spells. At African locations where a Chinese project had recently been completed and no other Chinese project remained active, the corresponding estimates are negative and statistically indistinguishable from zero. Under the baseline two-year post-completion window, the difference between ongoing and completed locations is not estimated precisely enough to reject equality; extending the window to three years makes the difference statistically significant for users and new spells. The completion evidence is therefore suggestive rather than decisive, but it points toward a connection created by project implementation rather than a permanent characteristic of the destination.

Taken together, the results are consistent with ongoing projects creating temporary placement capacity at specific African locations. A firm already operating at a project site has organizational infrastructure that can facilitate movement to that destination: recruitment contacts, accommodation, transport arrangements, administrative support, and an established presence on the ground. These arrangements perform some of the functions normally associated with migrant networks, but arise with project implementation rather than only through accumulated past migration. Development projects can therefore alter the connectivity of African locations to foreign labor markets even when they do not increase the overall volume of cross-border mobility.

The interpretation has important limits. We observe users' reported Chinese origins and their presence at African destinations, but generally not their occupations, employers, or reasons for travel. The estimates should therefore be interpreted as changes in observed Chinese mobility and presence rather than as changes in project-worker headcounts. Weibo captures only a selected share of Chinese mobility, recorded strikes are an incomplete measure of underlying labor-market deterioration, and a corridor-specific trend linking particular Chinese origins to firms expanding abroad cannot be fully ruled out.

The broader implication is that development finance can reshape the international connectivity of African locations as a by-product of project implementation. By determining which firms establish operations at particular sites, development projects create temporary links between recipient

locations and economic conditions abroad. These links affect not only where capital is deployed, but also where foreign mobility appears within Africa and how quickly it responds to changes in the economies to which implementing firms are connected. Development projects therefore shape both the geography of investment and the geography of cross-border economic integration in Africa.

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Table 1: Origin-destination reallocation after Chinese labour shocks

	Active in t	Active in $t - 1$ and t
Any user	0.33 (0.24)	0.39 (0.26)
Number of users	-0.12 (0.15)	-0.05 (0.15)
Posts	0.04 (0.34)	0.07 (0.37)
Any new spell	0.62** (0.26)	0.65** (0.28)
Number of new spells	0.05 (0.23)	0.17 (0.26)

Entries are percentage changes in an origin's expected destination-level outcome in project locations relative to non-project locations for one additional construction strike in the Chinese origin prefecture. PPML models include origin-destination-pair, origin-by-year, and destination-by-year fixed effects. Standard errors are clustered by Chinese origin prefecture.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A Numerical Estimates Corresponding to the Main Annual Figures

Tables 2 and 3 report the numerical estimates underlying Figures 1 and 3. Table 1 additionally reports the contemporaneous allocation estimates under the broader active-in- t definition.

Table 2: Annual lead test for origin-destination reallocation

Outcome	Active in $t - 1$	Same project in $t - 2, t - 1$	Strike year
Number of users	0.52** (0.21)	0.49** (0.24)	-0.05 (0.15)
Number of new spells	0.74*** (0.25)	0.70** (0.30)	0.17 (0.26)
Any user	0.52* (0.31)	0.32 (0.33)	0.39 (0.26)
Any new spell	0.46 (0.30)	0.24 (0.34)	0.65** (0.28)
Posts	0.72* (0.42)	0.64* (0.36)	0.07 (0.37)

Entries are percentage changes in an origin's expected destination-level outcome in project locations relative to other destinations for one additional construction strike in the Chinese origin prefecture. The first two columns use mobility in $t - 1$ and strikes in t and do not condition on project status in t . The second column requires the same project to be active in both $t - 2$ and $t - 1$. The strike-year column requires project activity in both $t - 1$ and t . PPML models cover the same 2014–2019 strike years and include origin-destination-pair, origin-by-year, and destination-by-year fixed effects. Standard errors are clustered by Chinese origin prefecture. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Dynamic Any-user response in ongoing-project locations

Horizon	Full-history shares	Leave-one-year-out shares
$h = -1$	-0.55 (0.48)	-0.25 (0.42)
$h = 0$	1.29** (0.63)	1.06* (0.54)
$h = 1$	1.00 (0.61)	1.02 (0.70)
$h = 2$	-0.49 (0.44)	-0.37 (0.33)
$h = 3$	-0.74 (0.96)	-0.56 (0.77)

Entries are the additional changes in Any-user presence in destinations with a project active in both $t - 1$ and t , in percentage points per unit of construction-strike exposure. At $h = -1$, the dependent variable is $Y_{t-1} - Y_{t-2}$. At $h = 0, \dots, 3$, it is $Y_{t+h} - Y_{t-1}$. Models include destination and country-by-base-year fixed effects. Standard errors are clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B Monthly Timing and Spell-Composition Diagnostics

Table 4 reports all jointly estimated monthly windows for the three country-spell outcomes. In the pre-strike row, the mobility outcome is measured one to three months before the strikes counted in that window. This is a falsification test for a pure response initiated by the strike and a diagnostic—but not an identified test—of anticipatory local adjustment.

Table 4: Monthly timing of strikes and mobility toward ongoing-project destinations

Timing	Construction strikes			All strikes		
	Users	Starts	Any start	Users	Starts	Any start
Pre-strike: 1–3 months	0.85** (0.41)	0.82** (0.40)	0.76* (0.44)	0.26 (0.32)	0.26 (0.31)	0.45 (0.31)
Same month	-0.38 (0.86)	-0.42 (0.87)	-0.21 (1.29)	-0.75 (0.65)	-0.76 (0.63)	-0.65 (0.79)
Past strikes: 1–3 months	0.49 (0.52)	0.47 (0.51)	0.92 (0.64)	0.01 (0.30)	-0.02 (0.31)	0.17 (0.31)
Past strikes: 4–6 months	-0.25 (0.42)	-0.25 (0.41)	-0.22 (0.48)	-0.04 (0.39)	-0.03 (0.39)	-0.14 (0.32)
Past strikes: 7–12 months	-0.50 (0.40)	-0.51 (0.41)	-0.52 (0.41)	-0.03 (0.21)	-0.04 (0.22)	-0.11 (0.25)

Entries are $100[\exp(\hat{\theta}) - 1]$: the percentage differential in the expected outcome in destinations with a project active in both $t - 1$ and t , per additional strike in the indicated window. Every window enters jointly. Pair, Chinese-origin-by-month, and African-destination-by-month fixed effects are included. Standard errors, in parentheses, are clustered by Chinese origin prefecture.

The outcome month runs from January 2014 through September 2019. In the first row, mobility precedes the strike by one to three months. It is a falsification test for a pure post-strike response and a diagnostic for possible anticipation. Coefficients for construction and all strikes are not directly comparable in magnitude because the event frequencies differ.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 4 and Table 5 compare Chinese and World Bank project locations in the same monthly PPML regressions. The Chinese pre-strike pattern remains visible conditional on World Bank activity. World Bank coefficients are instead concentrated in the first three months after a strike. The comparison is informative about financing footprints, but it is not a no-Chinese-personnel placebo because Chinese contractors can implement World Bank-financed projects.

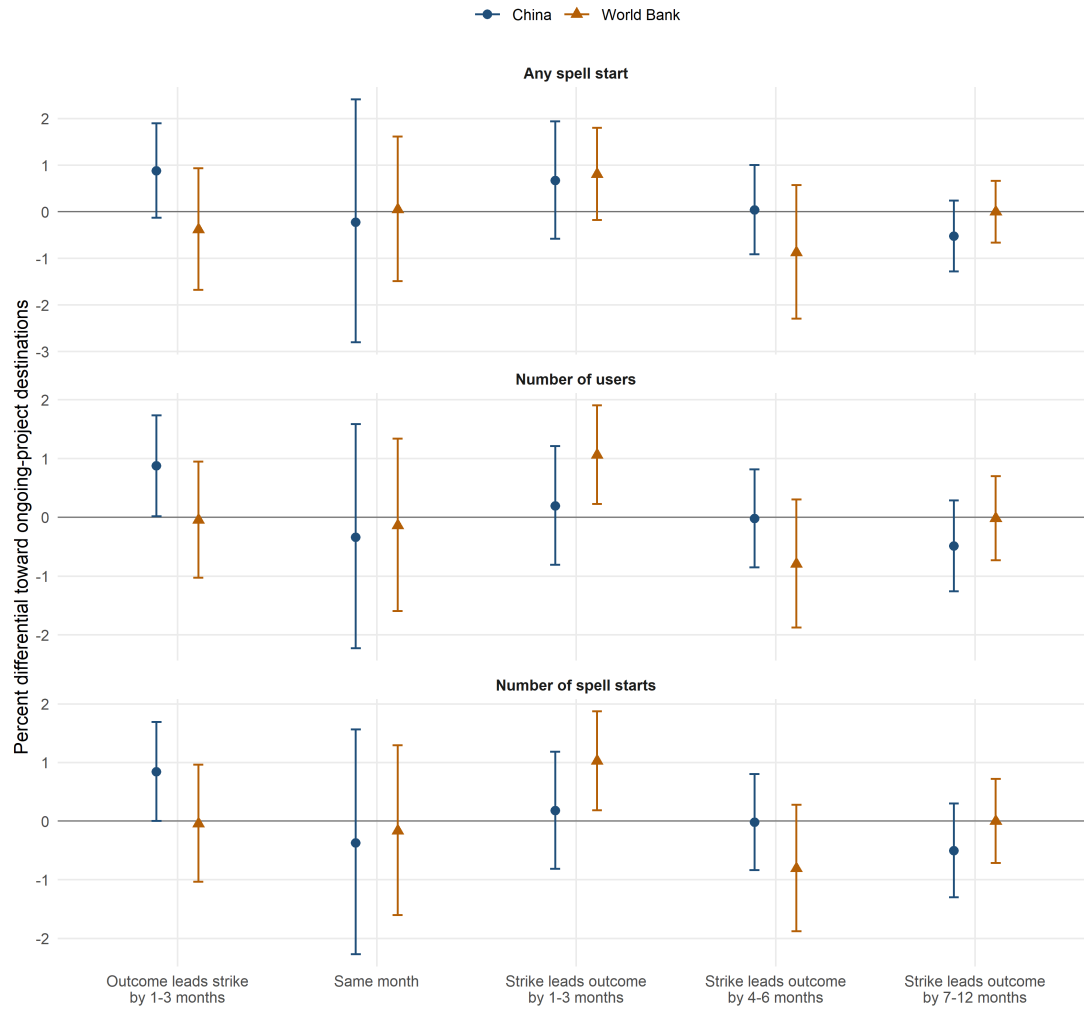


Figure 4: Monthly timing at Chinese and World Bank project locations

Notes: Points show percentage reallocations toward destinations with projects active in both the previous and current calendar year. Chinese- and World-Bank-project interactions for all five timing windows enter jointly. Bars are 95 percent confidence intervals based on standard errors clustered by Chinese origin prefecture.

Table 5: Monthly timing by project donor

Timing	Users		Starts		Any start	
	China	WB	China	WB	China	WB
<i>Construction strikes</i>						
Pre-strike: 1–3 months	0.87** (0.44)	-0.05 (0.50)	0.84** (0.43)	-0.04 (0.51)	0.88* (0.52)	-0.38 (0.67)
Same month	-0.34 (0.97)	-0.14 (0.75)	-0.37 (0.98)	-0.17 (0.74)	-0.23 (1.33)	0.05 (0.79)
Past strikes: 1–3 months	0.19 (0.52)	1.06** (0.43)	0.18 (0.51)	1.03** (0.43)	0.67 (0.64)	0.80 (0.50)
Past strikes: 4–6 months	-0.02 (0.43)	-0.79 (0.55)	-0.02 (0.42)	-0.81 (0.55)	0.04 (0.49)	-0.88 (0.73)
Past strikes: 7–12 months	-0.49 (0.39)	-0.02 (0.37)	-0.51 (0.41)	-0.00 (0.37)	-0.52 (0.39)	-0.01 (0.34)
<i>All strikes</i>						
Pre-strike: 1–3 months	0.19 (0.33)	0.21 (0.40)	0.19 (0.32)	0.25 (0.40)	0.32 (0.34)	0.43 (0.33)
Same month	-0.53 (0.75)	-0.70 (0.49)	-0.55 (0.73)	-0.70 (0.49)	-0.55 (0.87)	-0.30 (0.45)
Past strikes: 1–3 months	-0.20 (0.33)	0.71*** (0.26)	-0.22 (0.33)	0.67*** (0.25)	-0.09 (0.31)	0.88** (0.36)
Past strikes: 4–6 months	0.07 (0.36)	-0.36 (0.34)	0.09 (0.35)	-0.37 (0.34)	0.00 (0.36)	-0.43 (0.46)
Past strikes: 7–12 months	-0.06 (0.25)	0.15 (0.21)	-0.07 (0.26)	0.15 (0.22)	-0.06 (0.28)	-0.10 (0.22)

Entries are percentage reallocations toward destinations with projects active in both the previous and current calendar year. Chinese- and World-Bank-project interactions for all five timing windows enter jointly in each PPML model. Pair, Chinese-origin-by-month, and African-destination-by-month fixed effects are included. Standard errors are clustered by Chinese origin prefecture.

WB denotes World Bank. A World Bank-financed location is not a no-Chinese-involvement placebo because Chinese contractors can win and implement World Bank-financed civil-works contracts. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6 reports formal differences between spells that visit one harmonized administrative unit and spells that visit multiple units. The positive one-to-three-month contrasts are not sufficient to establish a worker response: the one-unit coefficients themselves are imprecise, and the contrast changes sign in the subsequent window.

Table 6: Monthly timing: one-unit versus multi-unit spell responses

Timing	Construction strikes			All strikes		
	Users	Starts	Any start	Users	Starts	Any start
Pre-strike: 1–3 months	3.32** (1.64)	3.27** (1.61)	1.86 (1.92)	1.17 (0.94)	1.17 (0.94)	1.15 (1.23)
Same month	-2.27 (2.41)	-2.28 (2.38)	-3.55 (2.65)	0.65 (1.67)	0.65 (1.65)	0.63 (1.69)
Past strikes: 1–3 months	2.78 (1.79)	2.78 (1.78)	3.57* (1.92)	2.12** (1.07)	2.11** (1.06)	2.69** (1.21)
Past strikes: 4–6 months	-4.27*** (1.19)	-4.28*** (1.21)	-2.84** (1.40)	-2.33*** (0.69)	-2.33*** (0.69)	-1.86*** (0.68)
Past strikes: 7–12 months	-0.54 (1.06)	-0.55 (1.06)	-1.26 (1.18)	-0.38 (0.62)	-0.39 (0.62)	-0.93 (0.61)

Entries are formal PPML contrasts between the percentage response of spells visiting one harmonized African administrative unit and spells visiting multiple units. They are behavioral proxies, not observed occupations. Positive entries indicate a larger response among one-unit spells. The model stacks both spell types and interacts each timing window with the one-unit indicator while allowing type-specific pair, origin-month, and destination-month fixed effects.

Standard errors, in parentheses, are clustered by Chinese origin prefecture. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C Donor Comparison and Sector Heterogeneity

Table 7 compares ongoing Chinese and World Bank project locations in the origin-destination PPML model. Both interactions enter jointly. The Chinese-project Any-new-spell coefficient remains positive and significant in the strike year, whereas the World Bank coefficient is near zero.

Table 7: Chinese- and World-Bank-project comparison in the origin-destination model

Project location	Year before strike ($t - 1$)	Strike year (t)
China	0.37 (0.31)	0.65** (0.29)
World Bank	0.51 (0.49)	0.01 (0.40)

Entries are percentage reallocations in the expected probability of an origin-destination cell recording a new spell per additional construction strike. Chinese- and World-Bank-project interactions enter jointly. In the strike-year column, projects must be active in both $t - 1$ and t . In the lead column, mobility and project activity are measured in $t - 1$ and strikes in t ; project activity in t is not imposed. Pair, origin-by-year, and destination-by-year fixed effects are included; standard errors are clustered by Chinese origin prefecture.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 5 and Table 8 report the corresponding donor comparison for the destination-level dynamic Any-user model. In contrast to the contemporaneous annual origin-destination result, World Bank locations exhibit positive differentials at the post-shock horizons, including when both donors enter together. This pattern is compatible with Chinese

participation in World Bank-financed implementation, but it does not distinguish that channel from characteristics common to major project locations.

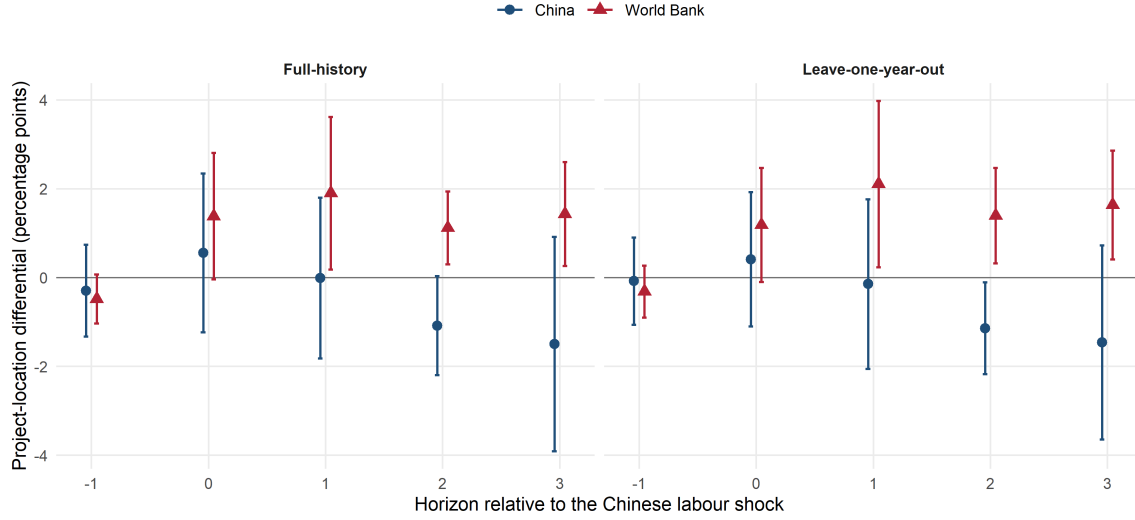


Figure 5: Chinese- and World-Bank-project comparison in the dynamic destination-level model
Notes: Points show the additional change in Any-user presence at ongoing Chinese or World Bank project locations per unit of construction-strike exposure. Both donor interactions enter jointly. At $h = -1$, the outcome is $Y_{t-1} - Y_{t-2}$; at $h = 0, \dots, 3$, it is $Y_{t+h} - Y_{t-1}$. Bars are 95 percent confidence intervals based on African-country clustered standard errors.

Table 8: Conditional dynamic Any-user donor-placebo estimates

Project location	$h = -1$	$h = 0$	$h = 1$	$h = 2$	$h = 3$
<i>Full-history shares</i>					
China	-0.30 (0.53)	0.56 (0.91)	-0.01 (0.92)	-1.08* (0.57)	-1.50 (1.23)
World Bank	-0.48* (0.28)	1.38* (0.73)	1.90** (0.88)	1.12*** (0.42)	1.43** (0.60)
<i>Leave-one-year-out shares</i>					
China	-0.08 (0.50)	0.41 (0.77)	-0.14 (0.97)	-1.14** (0.53)	-1.46 (1.12)
World Bank	-0.31 (0.30)	1.18* (0.65)	2.11** (0.95)	1.39** (0.55)	1.63** (0.62)

Entries are additional changes in Any-user presence, in percentage points per unit of construction-strike exposure. Chinese- and World-Bank-project interactions enter jointly in each destination-level model. At $h = -1$, the outcome is $Y_{t-1} - Y_{t-2}$; at $h = 0, \dots, 3$, it is $Y_{t+h} - Y_{t-1}$. Models include destination and country-by-base-year fixed effects. Standard errors are clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.1 Sector heterogeneity

Figure 6 and Table 9 split ongoing Chinese projects by sector. Physical and labor-intensive destinations display the clearest extensive-margin coefficients, but only the strike-year Any-user contrast is significant at the 5-percent level in the broad full-panel comparison. Direct comparisons restricted to project locations are imprecise.

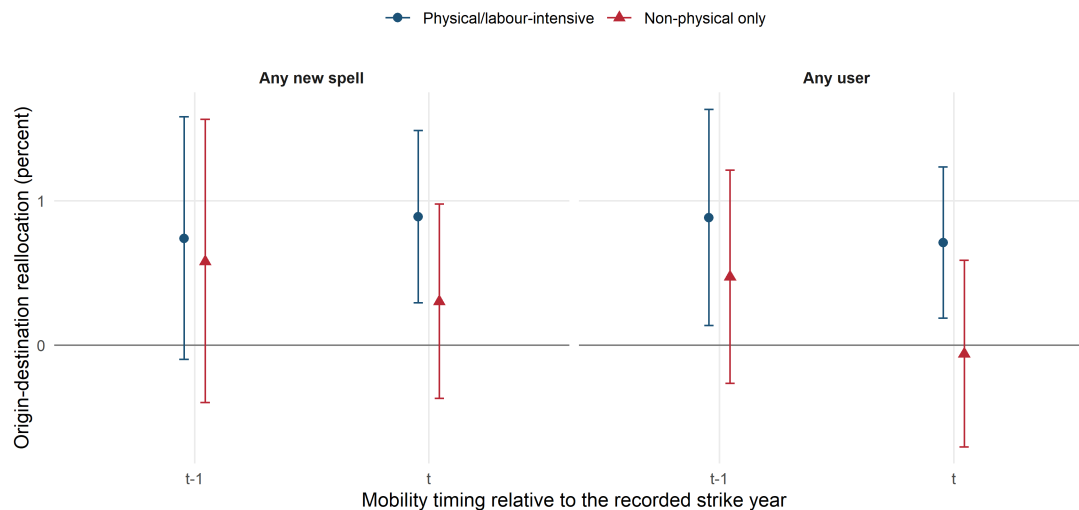


Figure 6: Sector heterogeneity in extensive-margin reallocation

Notes: Points show percentage changes in expected extensive-margin outcomes per additional construction strike for destination-years containing an ongoing physical/labor-intensive project or only non-physical ongoing projects. The two interactions enter jointly. Bars are 95 percent confidence intervals based on standard errors clustered by Chinese origin prefecture.

Table 9: Sector heterogeneity in origin-destination reallocation

	Year before strike ($t - 1$)	Strike year (t)
<i>Any new spell</i>		
Physical/labour-intensive projects	0.74* (0.43)	0.89*** (0.30)
Only non-physical projects	0.58 (0.50)	0.30 (0.34)
Physical minus non-physical	0.16 (0.60)	0.58 (0.40)
Physical minus non-physical, project sample	0.35 (0.52)	0.19 (0.44)
<i>Any user</i>		
Physical/labour-intensive projects	0.88** (0.38)	0.71*** (0.27)
Only non-physical projects	0.47 (0.38)	-0.06 (0.33)
Physical minus non-physical	0.41 (0.39)	0.77** (0.34)
Physical minus non-physical, project sample	0.38 (0.35)	0.33 (0.39)

Entries are percentage reallocations from jointly estimated PPML models. Physical/labour-intensive projects comprise transport, energy, water and sanitation, communications, and industry/mining/construction. The last row of each panel restricts the sample to destination-years with an exact project active in both years; other rows use the full origin-destination panel. All models include pair, origin-by-year, and destination-by-year fixed effects and cluster by Chinese origin prefecture.

The analysis is exploratory and reports unadjusted p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D Project-Start Difference-in-Differences

This appendix reports the earlier project-start design. Figure 7 shows event-study estimates for the Any-user outcome, with event time -1 as the reference period. Table 10 summarizes average post-treatment effects for the extensive-margin user and spell outcomes.

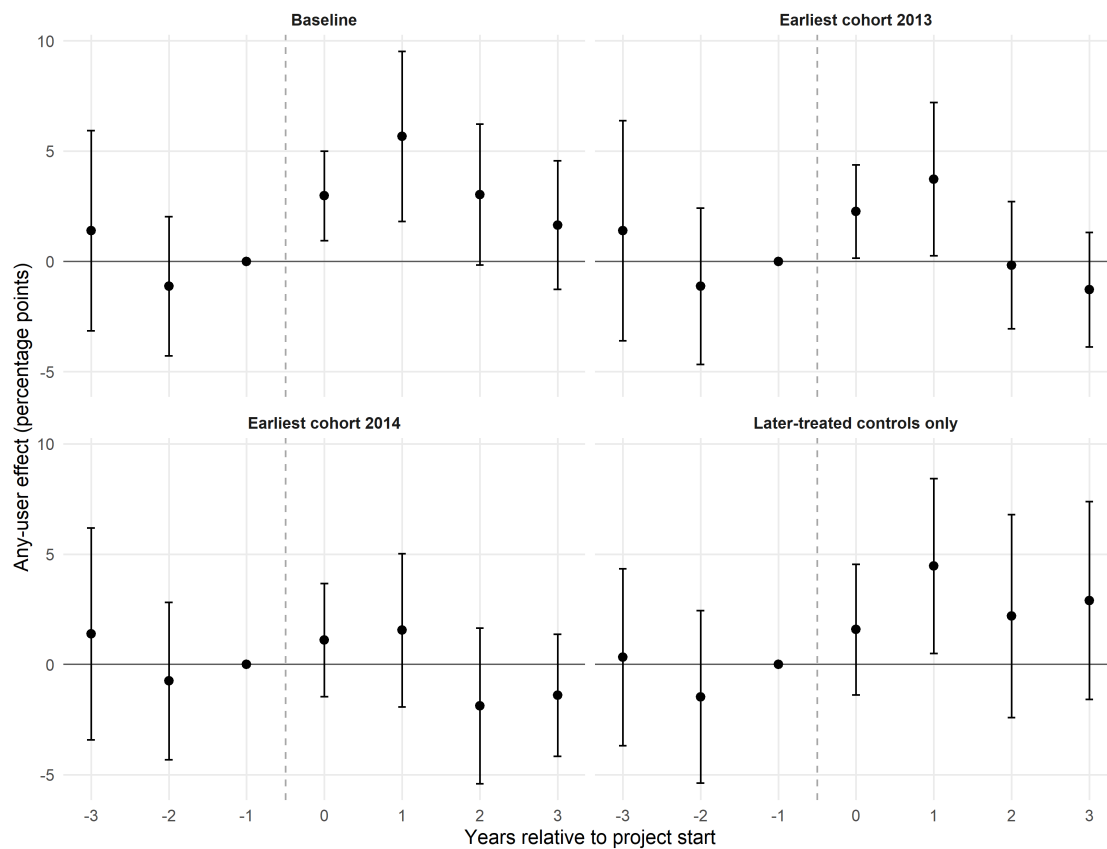


Figure 7: Project-start event studies for the Any-user outcome

Notes: Callaway-Sant’Anna group-time average treatment effects aggregated by event time. Event time -1 is the reference category. The panels show the baseline sample, two later-cohort restrictions, and the later-treated-only comparison. Confidence intervals use 999 bootstrap repetitions.

Table 10: Project-start CS-DID estimates

Specification	Average ATT, event times 0–3
Baseline: any user	3.330*** (1.192)
Baseline: any new spell	3.654*** (1.087)
World Bank placebo: any user	0.435 (0.714)
World Bank placebo: any new spell	0.164 (0.611)
Persistent spells: any new spell	3.523*** (1.084)
Strict spells: any new spell	2.839*** (0.958)
Earliest cohort 2013: any user	1.133 (1.006)
Earliest cohort 2013: any new spell	1.869** (0.932)
Earliest cohort 2014: any user	-0.157 (1.149)
Earliest cohort 2014: any new spell	0.845 (1.054)
Later-treated only: any user	2.784* (1.500)
Later-treated only: any new spell	3.852*** (1.353)

Estimates are percentage points. Event time -1 is the universal reference period.

Controls are not-yet-treated and never-treated units unless the row states otherwise.

Inference uses 999 multiplier-bootstrap iterations clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

In the baseline sample, the average post-start effect over event times zero to three is 3.33 percentage points for Any user ($p = 0.005$) and 3.65 percentage points for Any new spell ($p < 0.001$). The corresponding World Bank estimates are small and imprecise. Alternative spell definitions also yield positive average effects.

However, excluding the earliest project cohorts substantially weakens the Any-user result. With 2013 as the earliest cohort it is 1.13 percentage points and insignificant, and with 2014 as the earliest cohort it is -0.16 percentage points and insignificant. The spell outcome is somewhat more stable, but its magnitude also changes. Because early project cohorts overlap with the weakest platform coverage and provide short pre-periods, these estimates cannot cleanly separate project activation from changes in observability. They remain useful descriptive context but are not the basis for the paper’s central claim.

E Longer Sample and the Broad Strike Measure

Table 11 reports the complete 2011–2019 comparison requested for the earlier reduced-form window. It crosses construction-only and all-strike shocks with full-history and leave-one-year-out shares. The table reports both the

interaction and the linear combination giving the total exposure response in active-project units.

Table 11: Reduced-form estimates in the 2011-2019 sample

	Construction strikes		All strikes	
	Full-history	Leave-one-year-out	Full-history	Leave-one-year-out
<i>Panel A: Any user</i>				
Active-project differential	2.20** (0.99)	1.86** (0.84)	0.93** (0.35)	0.70** (0.28)
Total effect in active units	3.08*** (0.97)	2.68*** (0.83)	1.30*** (0.34)	1.06*** (0.28)
<i>Panel B: Any new spell</i>				
Active-project differential	1.43 (0.89)	1.24 (0.77)	0.66* (0.34)	0.49* (0.27)
Total effect in active units	2.47*** (0.84)	2.20*** (0.73)	1.12*** (0.32)	0.94*** (0.26)

Coefficients and standard errors are in percentage points per one-unit increase in the indicated exposure. Because all-strike and construction-strike exposures have different distributions, their coefficient magnitudes are not directly comparable. All models include destination and country-by-year fixed effects and cluster by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The Any-user interaction remains positive and significant under every construction. Using all strikes therefore reinforces rather than overturns the longer-window extensive-margin result. The Any-new-spell interaction is less precise, but the total effect in active units remains positive and significant. Because one unit of the all-strike exposure represents a different and more common shock than one unit of construction-strike exposure, coefficient levels should not be compared as if they were measured on the same scale.

F Exposure-Tail and Influence Diagnostics

Figure 8 displays the upper-tail checks, and Table 12 adds leave-one-country-out and leave-one-origin-out summaries. Capping the exposure changes the metric attached to a one-unit increase, which explains why the coefficient can rise when the maximum is compressed. Dropping observations instead preserves the original scale but reduces identifying variation and, at aggressive cutoffs, precision.

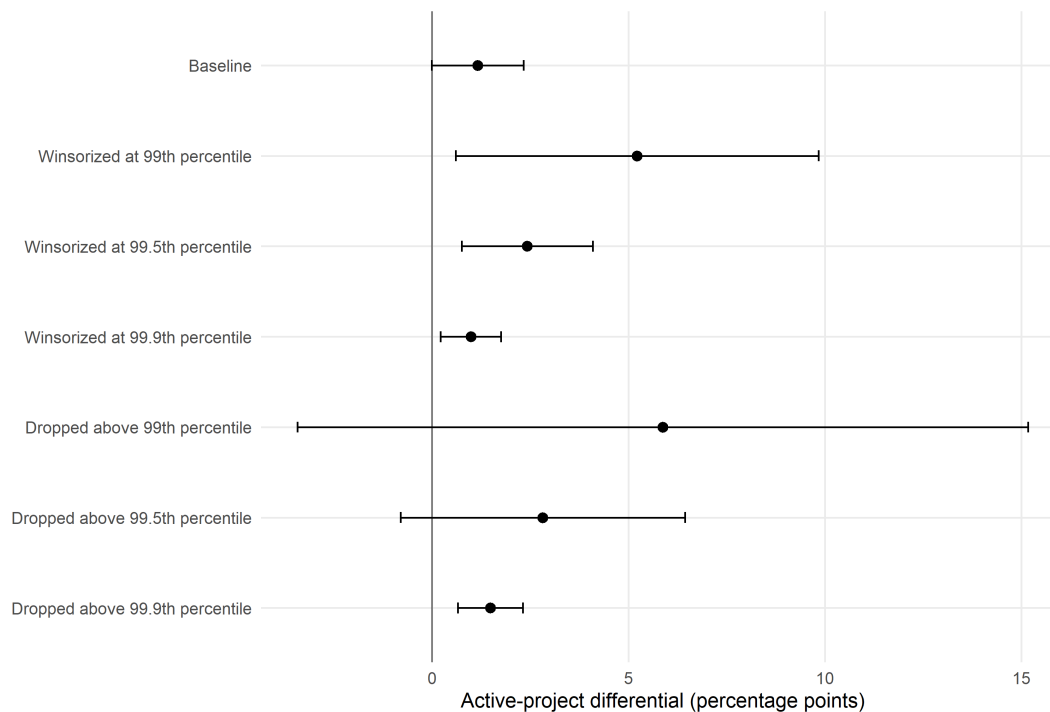


Figure 8: Sensitivity to the upper tail of construction-strike exposure

Notes: Points show the active-project differential in the 2014–2019 Any-user model. “Winsorized” specifications cap exposure at the indicated percentile; “dropped” specifications remove observations above it. Bars are 95 percent confidence intervals based on African-country clustered standard errors.

Table 12: Exposure-tail and influence diagnostics

Diagnostic	Differential	Standard error	Observations
Baseline	1.16*	(0.58)	39,420
Winsorized at 99th percentile	5.22**	(2.30)	39,420
Winsorized at 99.5th percentile	2.42***	(0.83)	39,420
Winsorized at 99.9th percentile	0.99**	(0.38)	39,420
Dropped above 99th percentile	5.87	(4.64)	38,998
Dropped above 99.5th percentile	2.82	(1.80)	39,213
Dropped above 99.9th percentile	1.48***	(0.41)	39,379
Leave-one-African-country-out range	0.77 to 1.86		–
Leave-one-Chinese-origin-out range	1.13 to 1.21		–
Share positive: country / origin omissions	100% / 100%		–
Share with $p < 0.10$: country / origin omissions	98% / 100%		–

The first panel re-estimates the 2014–2019 Any-user model after capping or dropping the upper tail of the construction-strike exposure. The remaining rows summarize 54 leave-one-African-country-out and 295 leave-one-Chinese-origin-out regressions. Main inference is clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

G Origin-Shock and Clustering Diagnostics

Table 13 compares the conservative African-country clustered standard errors with the complementary origin-shock score calculation. Table 14 separately compares alternative destination-side clustering choices.

Table 13: Alternative inference for the Any-user differential

	Static reduced form	Dynamic DDD, $h = 0$
Point estimate	1.16*	1.68**
SE clustered by African country	(0.58)	(0.81)
Origin-shock score SE	(0.10)	(0.12)

Coefficients are in percentage points. Main standard errors allow arbitrary dependence within African countries. The origin-shock score calculation decomposes the shift-share regressor into its 276 nonzero Chinese-prefecture contributions and clusters those scores by origin prefecture. It is reported as a complementary shock-side diagnostic, not as a replacement for the conservative destination-country inference.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ based on the country-clustered inference shown in the first row.

Table 14: Destination-side clustering sensitivity

Clustering	Static differential	Dynamic differential, $h = 0$
African country	1.157* (0.584)	1.683** (0.809)
Administrative unit	1.157** (0.544)	1.683** (0.794)
Two-way	1.157* (0.584)	1.683** (0.809)

Estimates are percentage points. Two-way clustering combines African country and administrative unit; because units are nested within countries, country clustering is the binding component here. These destination-side checks do not replace formal shift-share shock-level inference.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

H Additional Dynamic Estimates

Table 15 compares the ongoing-project estimates with the broader active-in- t definition. Table 16 adds the Any-new-spell outcome under the broader definition.

Table 15: Dynamic labour-shock response of local Chinese presence

Horizon	Full-history shares		Leave-one-year-out	Ongoing project
	Differential	Total in active units	Differential	Differential
$h = 0$	1.683** (0.809)	1.688* (0.990)	1.395** (0.687)	1.293** (0.630)
$h = 1$	1.031 (0.643)	1.054 (0.823)	1.073 (0.716)	0.998 (0.611)
$h = 2$	-0.125 (0.393)	-0.327 (0.549)	-0.062 (0.349)	-0.487 (0.438)
$h = 3$	-0.462 (0.729)	-0.611 (0.936)	-0.319 (0.609)	-0.736 (0.959)

The dependent variable is the change in any-user presence from $t - 1$ to $t + h$. Estimates are percentage points per unit of construction-strike exposure. The ongoing-project definition requires activity in both $t - 1$ and t . All models include administrative-unit and country-by-base-year fixed effects. Standard errors are clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 16: Dynamic differential responses at the extensive margin

	$h = -1$	$h = 0$	$h = 1$	$h = 2$	$h = 3$
Any user	-0.54 (0.58)	1.68** (0.81)	1.03 (0.64)	-0.12 (0.39)	-0.46 (0.73)
Any new spell	—	0.84 (0.55)	0.50 (0.34)	0.04 (0.20)	-0.30 (0.44)

Entries are active-project interaction coefficients in percentage points. For $h = -1$, the dependent variable is the pre-shock change from $t - 2$ to $t - 1$; for $h = 0, \dots, 3$, it compares the outcome in $t + h$ with its value in $t - 1$. The Any-new-spell lead is not estimated in this table. Models include destination and country-by-year fixed effects; standard errors are clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

I Static Levels, Sample Sensitivity, and Additional Outcomes

Table 17 reports the static Any-user specification using the ongoing-project definition. Table 18 gives the corresponding broader active-in- t estimates. These coefficients describe conditional associations in outcome levels rather than changes from the preceding year.

Table 17: Static Any-user association using ongoing projects

	Full-history shares	Leave-one-year-out shares
Outside ongoing-project locations	0.23 (0.18)	0.05 (0.08)
Ongoing-project differential	0.89* (0.47)	0.59** (0.29)
Total effect in ongoing-project locations	1.12** (0.53)	0.64* (0.33)

Entries are percentage-point associations for Any-user presence. Ongoing projects are active in both $t - 1$ and t . Models cover 2014–2019 and include destination and country-by-year fixed effects. Standard errors are clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 18: Labour-shock exposure and local Chinese presence

	Full-history shares	Leave-one-year-out shares
Outside active-project units	0.217 (0.161)	0.035 (0.081)
Active-project differential	1.157* (0.584)	0.788** (0.390)
Total effect in active-project units	1.374** (0.655)	0.824* (0.428)
Observations	39,420	39,420
ADM-unit fixed effects	Yes	Yes
Country-by-year fixed effects	Yes	Yes

Notes: The dependent variable equals one if any sampled Chinese Weibo user is observed in the destination-year. Coefficients are percentage-point changes per one-unit increase in the shift-share exposure. The active-project differential is the interaction coefficient; the total effect is the sum of the exposure and interaction coefficients. Standard errors in parentheses are clustered by African country. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 9 and Table 19 vary the first sample year in the broader static specification. Precision is non-monotone across windows, which is one reason the static model is not the headline design.

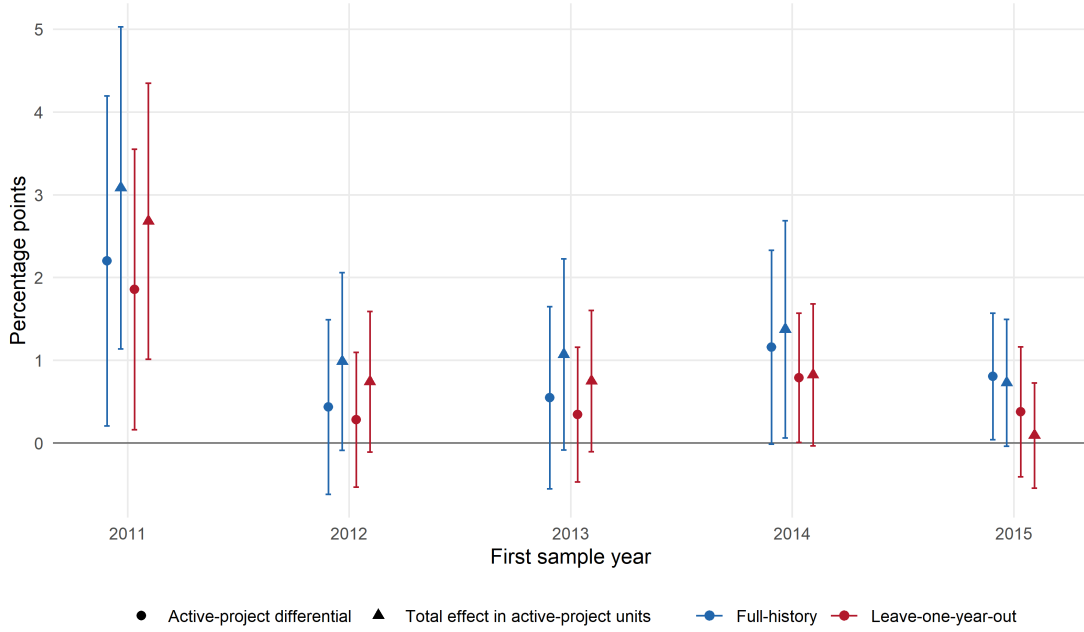


Figure 9: Sensitivity of the static active-project association to the sample start

Notes: Points show the interaction between construction-strike exposure and active-project status, varying the first sample year and retaining 2019 as the final year. Bars are 95 percent confidence intervals based on African-country clustered standard errors and finite-cluster t critical values.

Table 19: Sensitivity to the first sample year

First year	Full-history shares	Leave-one-year-out shares
2011	2.202** (0.994)	1.855** (0.844)
2012	0.436 (0.526)	0.283 (0.406)
2013	0.547 (0.549)	0.343 (0.407)
2014	1.157* (0.584)	0.788** (0.390)
2015	0.804** (0.381)	0.376 (0.391)

Entries are active-project differential effects in percentage points. Every window ends in 2019. Models include administrative-unit and country-by-year fixed effects; standard errors are clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 20 reports static conditional associations in outcome levels. The first column is the exposure slope in unit-years without an active project, the second is the difference in that slope when a project is active, and the third is the resulting slope in active-project unit-years. These are not changes from the preceding year.

Table 20: Outcome and exposure robustness

Specification	Non-active units	Active-project units	
	Exposure effect	Differential	Total effect
Any user	0.217 (0.161)	1.157* (0.584)	1.374** (0.655)
All strikes: Any user	0.004 (0.025)	0.358** (0.135)	0.363** (0.146)
Coverage-standardized: Any user	0.192 (0.145)	1.040* (0.531)	1.232** (0.599)
Users per 10,000 archive users	1.529*** (0.310)	-2.332*** (0.445)	-0.804* (0.401)
Number users	11.054*** (2.294)	-15.313*** (3.408)	-4.259 (3.095)
Posts	23.889*** (1.849)	-43.675*** (9.149)	-19.786** (9.261)
Any new spell	0.184 (0.141)	0.488 (0.418)	0.672 (0.464)
Number new spells	12.978*** (3.018)	-19.664*** (2.659)	-6.686*** (2.037)

The exposure effect is the response to labour-shock exposure in unit-years without an active project. The differential is the additional response in active-project unit-years, and the total effect is their sum. Binary outcomes and the coverage-standardized probability are in percentage points. Counts and rates retain the units named in the row. Coefficients using all strikes are not directly comparable in magnitude to construction-strike coefficients because the exposure scale differs. All models cover 2014–2019 and include administrative-unit and country-by-year fixed effects. Standard errors are clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 21 reports the corresponding immediate changes for the count outcomes. The negative estimates and their disagreement with the near-zero origin-destination PPML estimates are why the paper does not make a claim about an increase or a known decrease in migrant volumes.

Table 21: Immediate dynamic responses of count outcomes

Outcome	Outside active units	Differential	Total in active units
Number of users	-7.70** (2.93)	-2.56 (3.94)	-10.26*** (2.85)
Posts	-3.99 (6.60)	-24.23* (12.51)	-28.22*** (10.06)
Number of new spells	-9.72*** (2.42)	-1.42 (2.48)	-11.14*** (1.47)

The dependent variable is the change from $t - 1$ to t ($h = 0$). The differential is the additional response in active-project locations; the total is the sum of the outside slope and differential. Models cover 2014–2019 and include administrative-unit and country-by-year fixed effects. Standard errors are clustered by African country.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

J Total Origin Mobility and Historical Links

Table 22 tests whether strikes predict more observed mobility from the affected Chinese prefecture to Africa as a whole.

Table 22: Do strikes increase total observed Africa mobility?

Outcome	2014–2019		2011–2019
	Construction	All strikes	Construction
Users	0.28 (0.49)	-0.18 (0.30)	0.08 (0.48)
Posts	0.85** (0.41)	-0.19 (0.29)	0.78* (0.42)
Spells	0.37 (0.40)	-0.19 (0.31)	0.26 (0.39)
Destinations with a user	-0.26 (0.25)	-0.00 (0.19)	-1.36*** (0.36)
First-observed user destinations	-2.20*** (0.62)	-0.28 (0.49)	-4.71*** (0.86)

Entries are PPML percentage changes in the total outcome observed across Africa for one additional strike in a Chinese origin prefecture. Models include origin-prefecture and year fixed effects and cluster standard errors by origin prefecture. A first-observed destination is an origin-destination pair appearing for the first time since 2011.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

J.1 Exposure distribution

The standard construction-strike exposure is highly skewed, as is common when a small number of origin shocks receive large destination weights. In 2014–2019 its mean is 0.096, its 95th percentile is 0.071, its 99th percentile is 1.47, and its maximum is 163.23. The mean exceeds the 95th percentile because it is pulled upward by a very small number of extreme exposure values in the upper tail. The coefficient in the main text is the effect of a one-unit increase, so readers should not interpret that unit as a typical annual change for every destination. The leave-one-year-out shares, all-strike exposure, and sample-window sensitivity show that the main sign is not unique to one construction, although the long tail remains a reason for cautious interpretation.